

GLaRef@CRAC2025: Should We Transform Coreference Resolution into a Text Generation Task?

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Lattice (CNRS – École Normale Supérieure – Université Sorbonne Nouvelle), Paris, France

Shared Task Overview

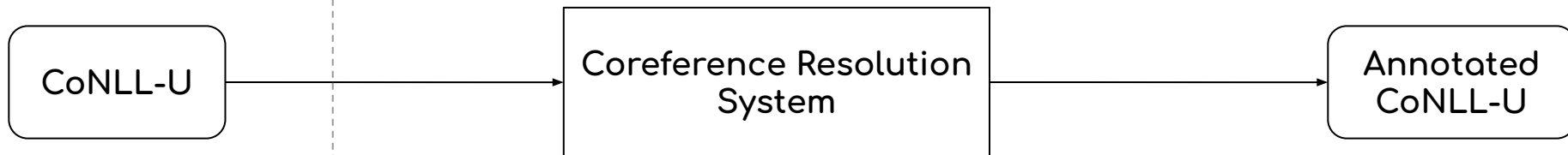


- CorefUD 1.3
- 17 Languages :
Ancient Greek, Biblical Hebrew, Catalan, Czech, English, French, German, Hindi, Hungarian, Korean, Lithuanian, Norwegian, Old Church Slavonic, Polish, Russian, Spanish, Turkish
- 22 Datasets
 - ↳ Documents
 - ↳ Sentences

Shared Task Overview: Overview

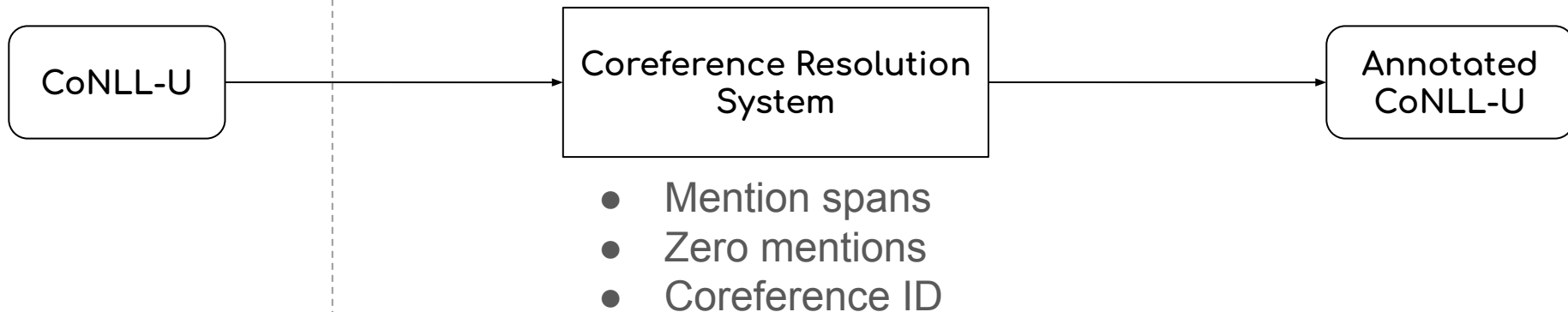


22 Datasets



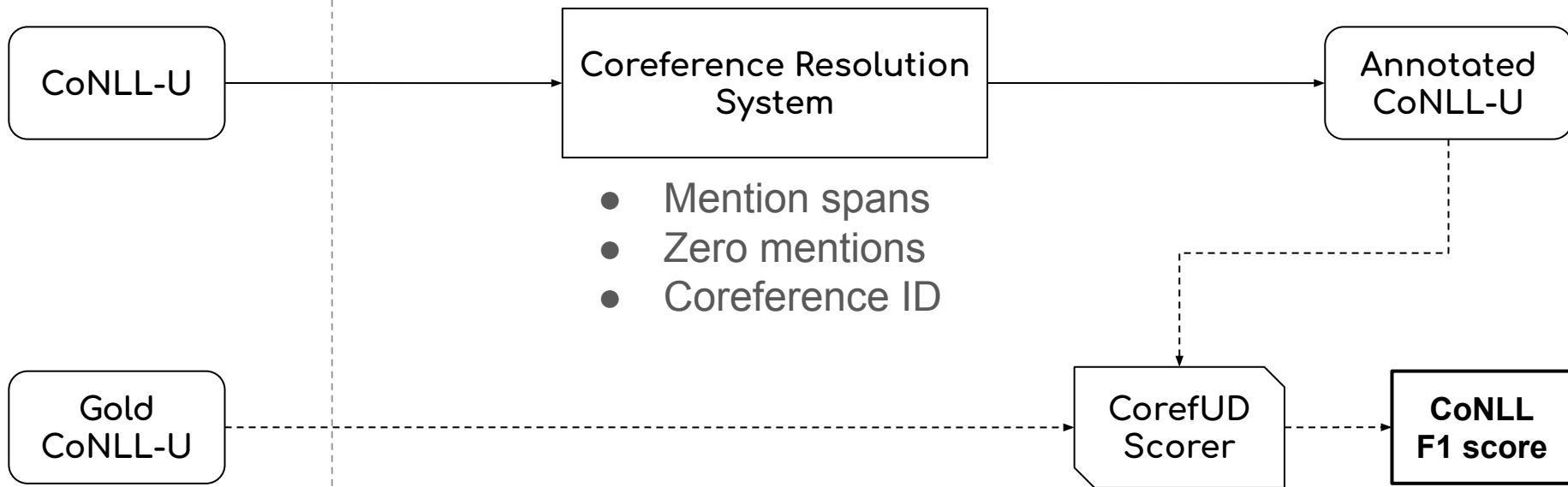
Shared Task Overview: Overview

22 Datasets

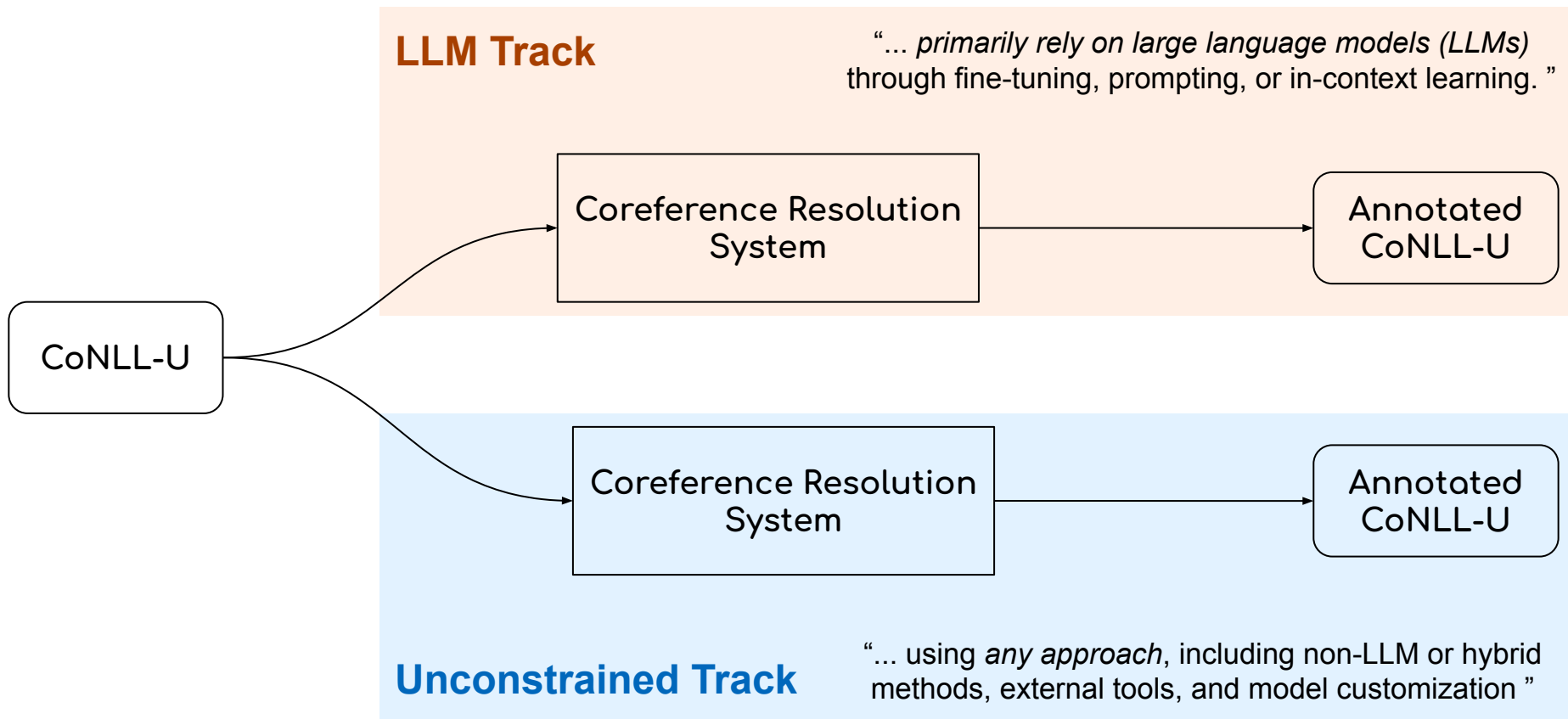


Shared Task Overview: Evaluation

22 Datasets



Unconstrained & LLM Tracks



Unconstrained Track

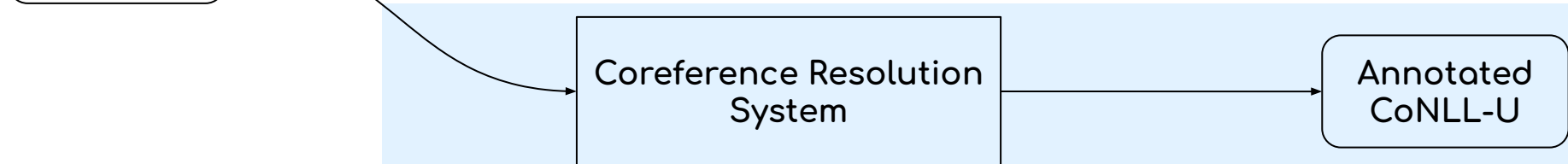
LLM Track

“... primarily rely on large language models (LLMs) through fine-tuning, prompting, or in-context learning.”



Unconstrained Track

*“... using **any approach**, including non-LLM or hybrid methods, external tools, and model customization ”*



Unconstrained Submission: Multistage Pipeline

CoNLL-U



The diagram illustrates a multistage pipeline for NLP tasks. It begins with a box labeled 'CoNLL-U' in the top left. A solid arrow points down from this box to the first stage, 'Token Embeddings'. This stage is separated from the next, 'Mentions Detection', by a vertical dashed line. The 'Mentions Detection' stage is further separated from 'Mention Pairs' by another vertical dashed line. Finally, 'Mention Pairs' is separated from the last stage, 'Clustering', by a third vertical dashed line. All stages are represented by text labels at the bottom of the diagram.



Token
Embeddings

Mentions Detection

Mention Pairs

Clustering

Unconstrained Submission: Multistage Pipeline

CoNLL-U



Multilingual
encoder
mT5-xl

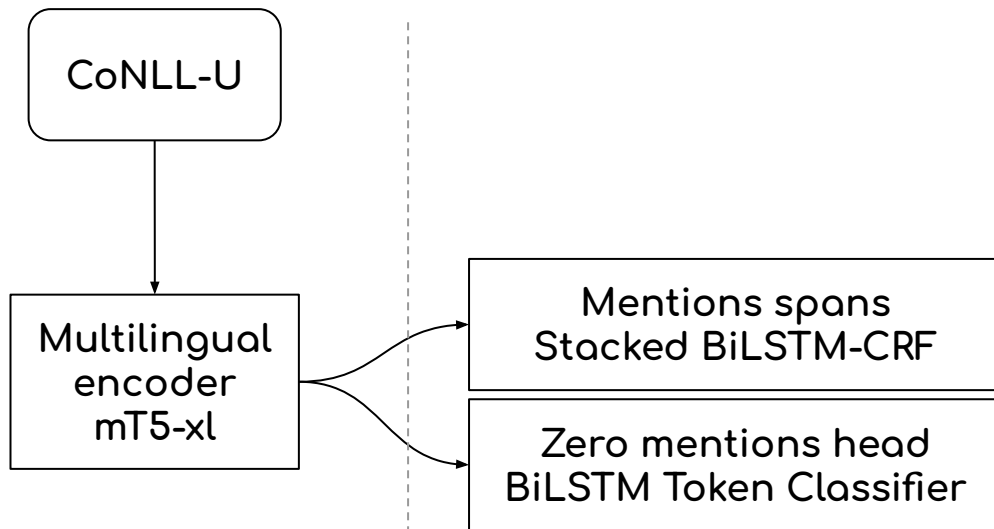
Token
Embeddings

Mentions Detection

Mention Pairs

Clustering

Unconstrained Submission: Multistage Pipeline



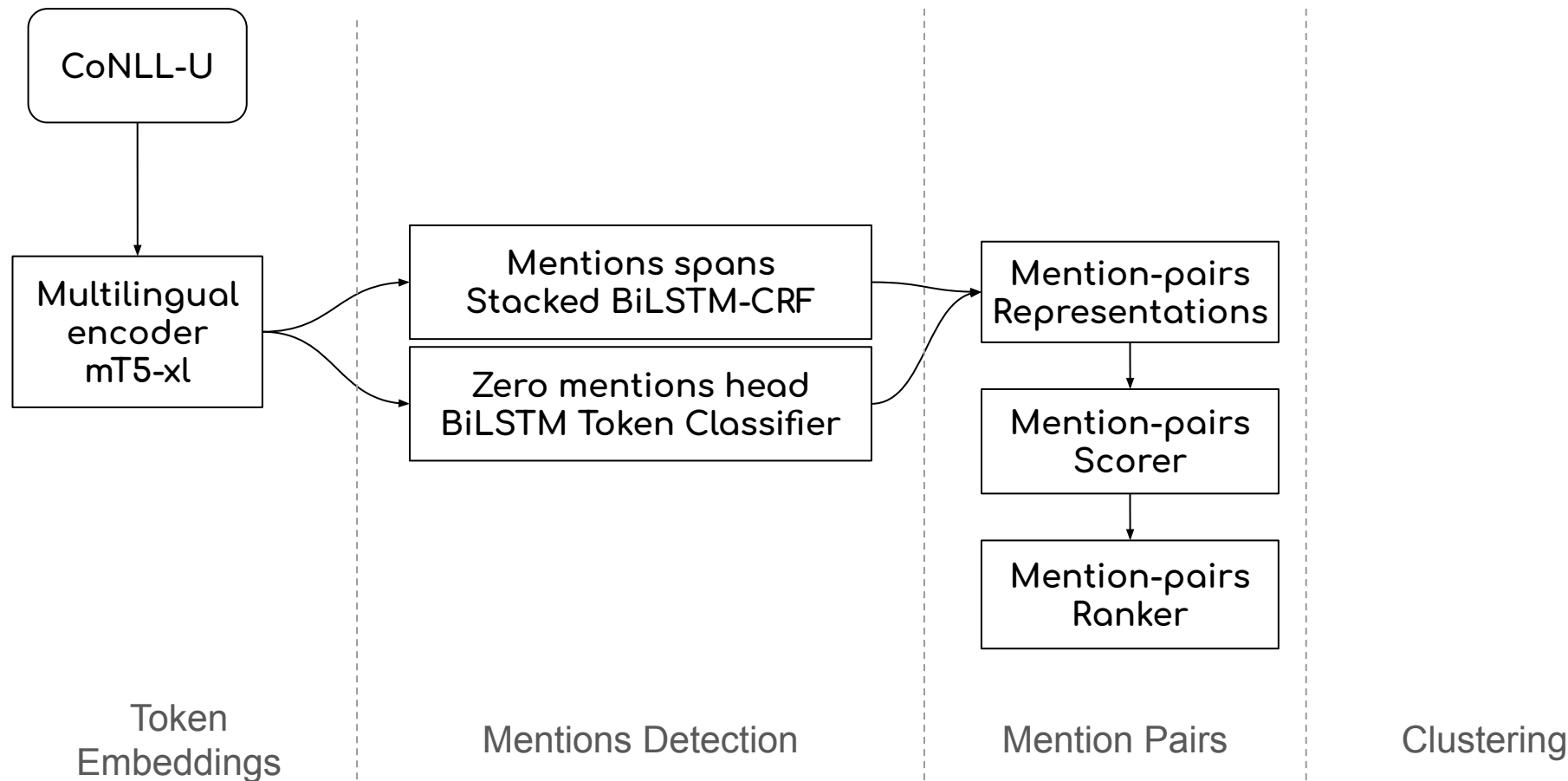
Token
Embeddings

Mentions Detection

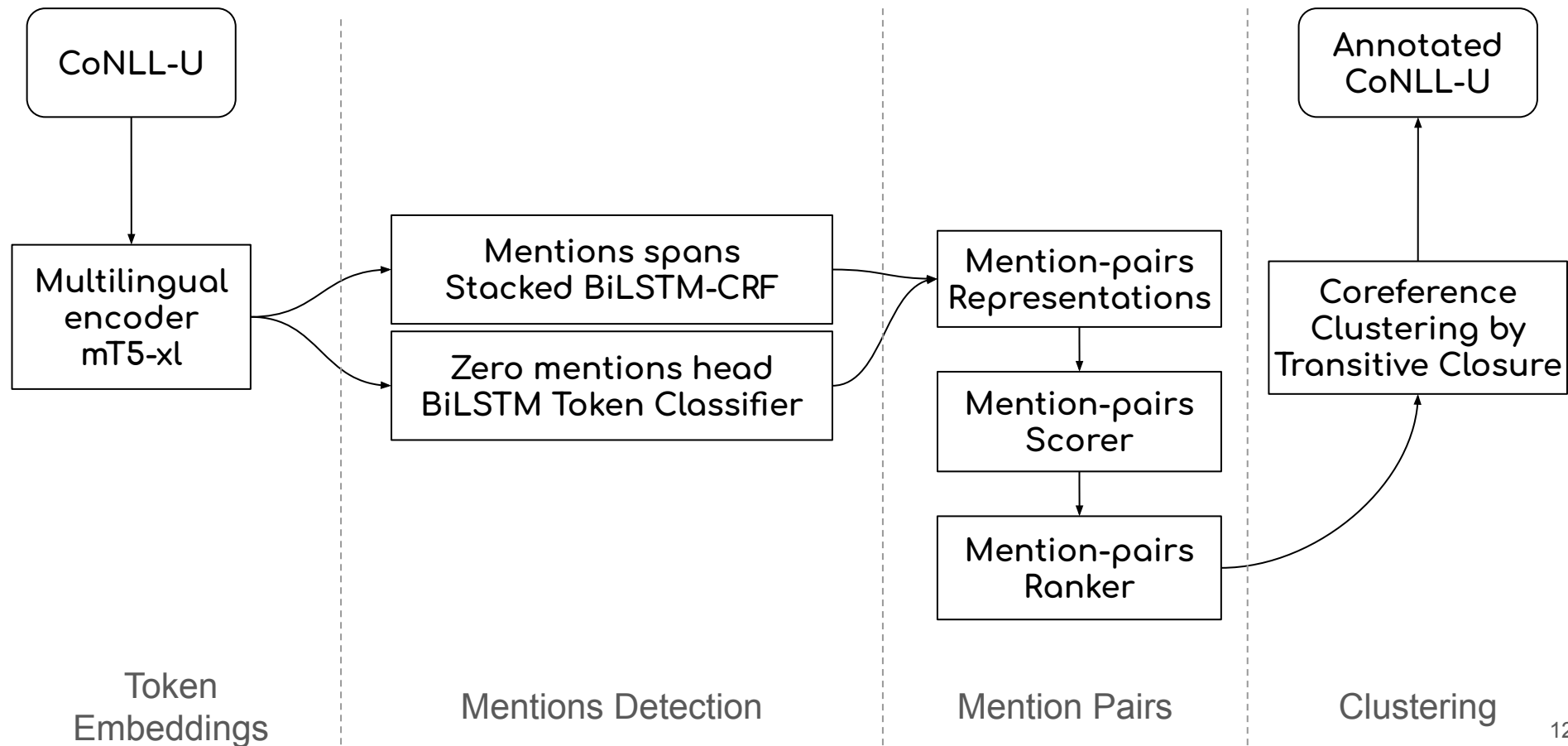
Mention Pairs

Clustering

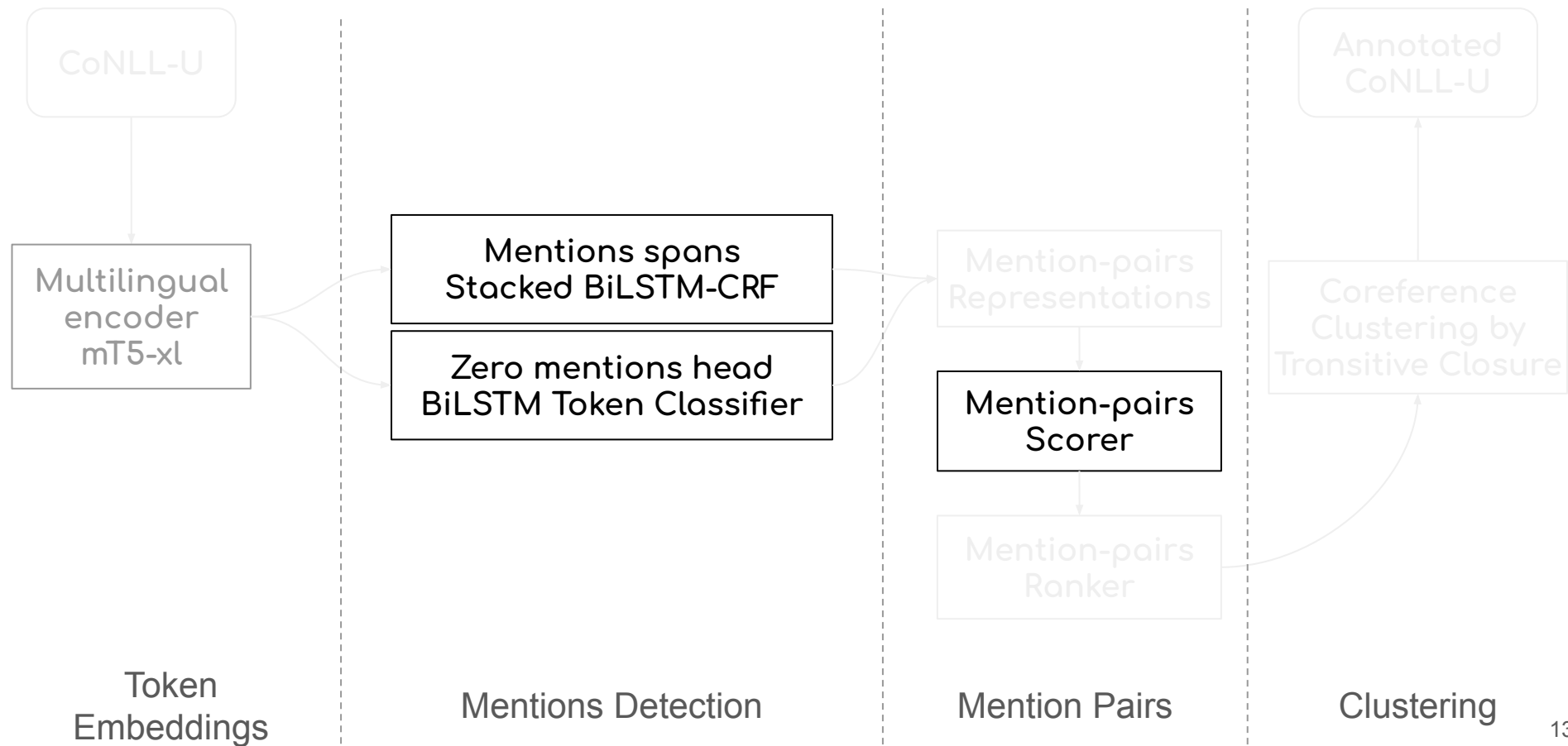
Unconstrained Submission: Multistage Pipeline



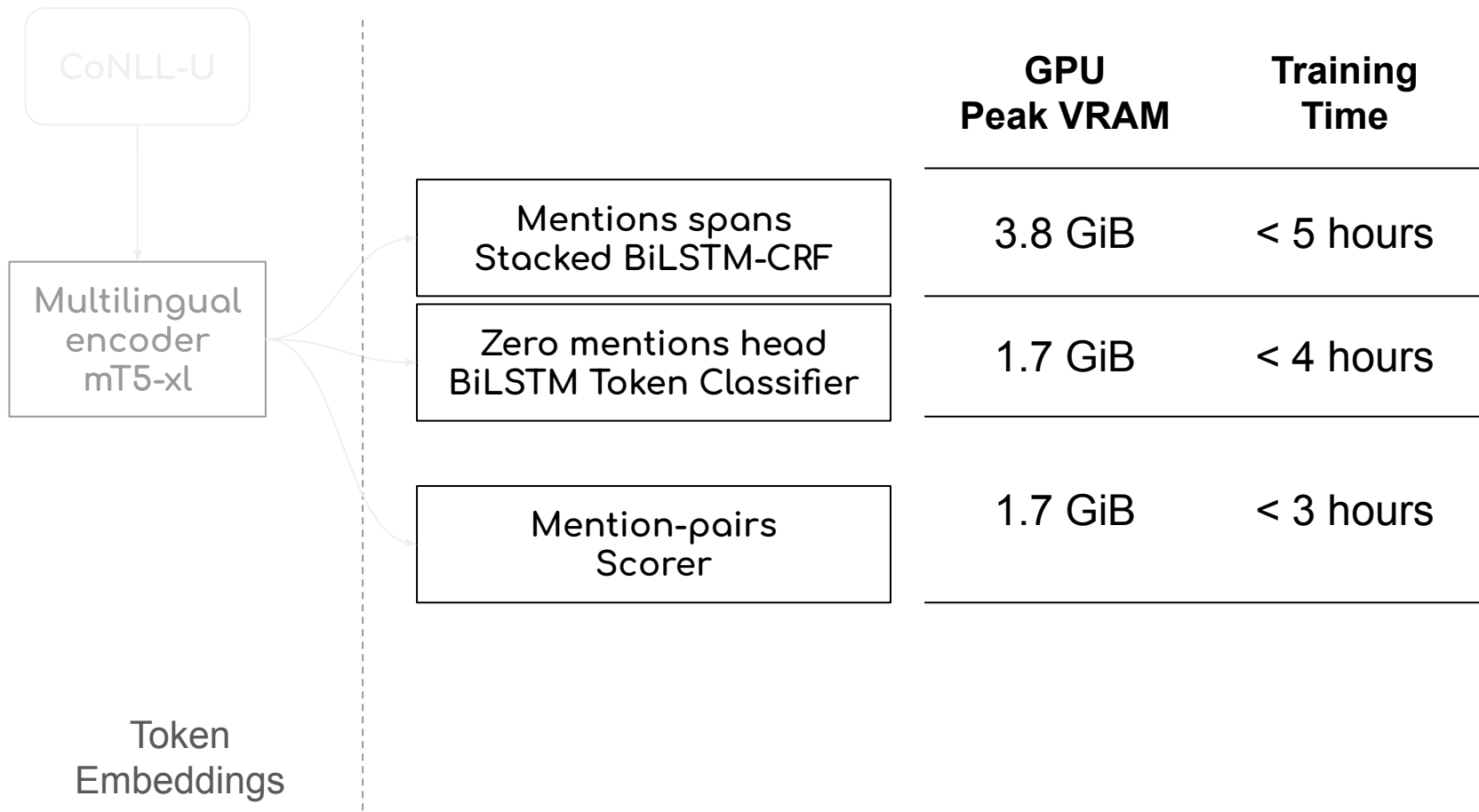
Unconstrained Submission: Multistage Pipeline



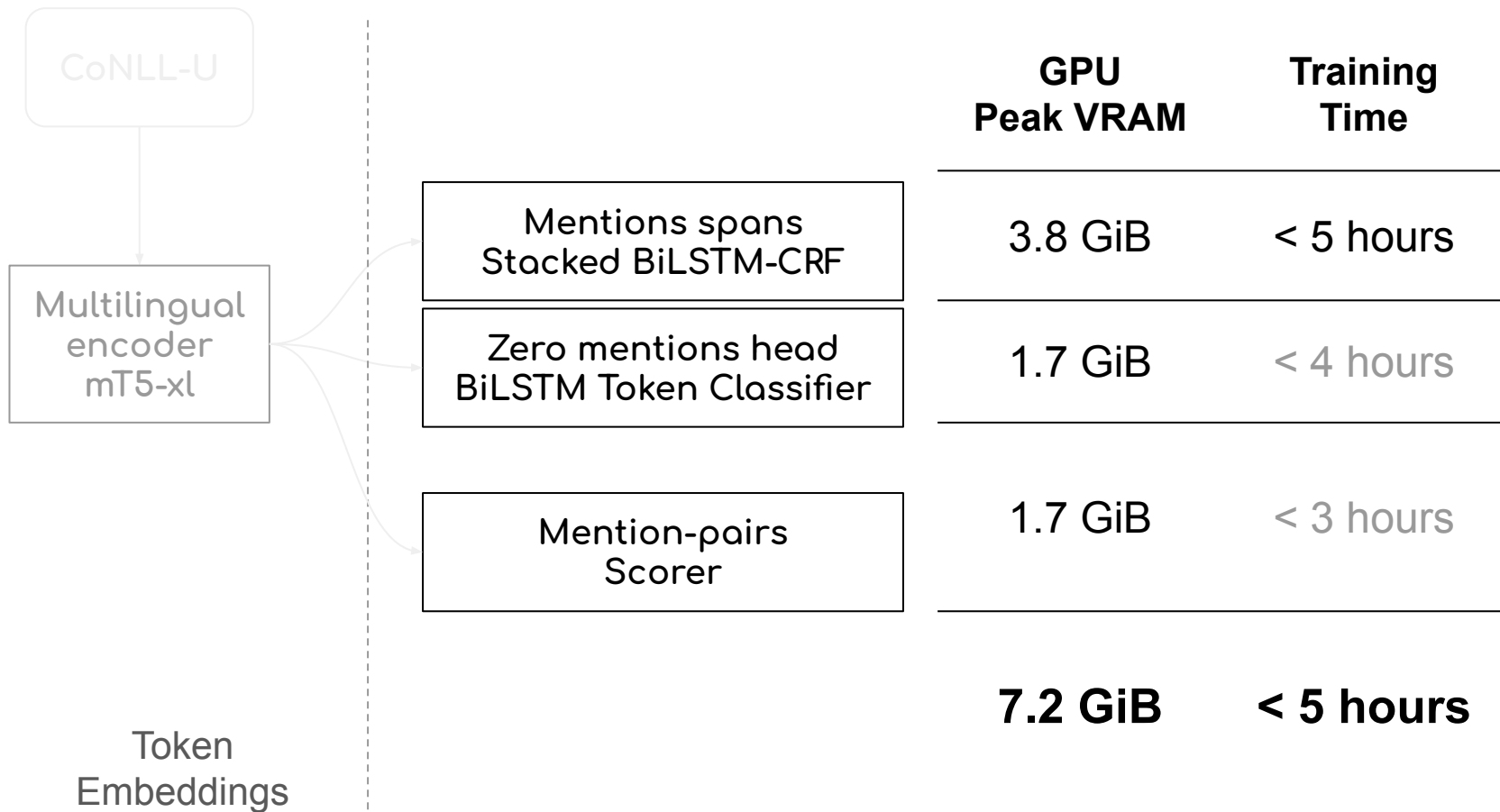
Unconstrained Submission: Trained Modules



Unconstrained Submission: Training Resources



Unconstrained Submission: Training Resources



Unconstrained Submission: Training Resources

CoNLL-U

Multilingual
encoder
mT5-xl

Mentions spans
Stacked BiLSTM-CRF

Zero mentions head
BiLSTM Token Classifier

Mention-pairs
Scorer

**GPU
Peak VRAM**

**Training
Time**

3.8 GiB

< 5 hours

1.7 GiB

< 4 hours

1.7 GiB

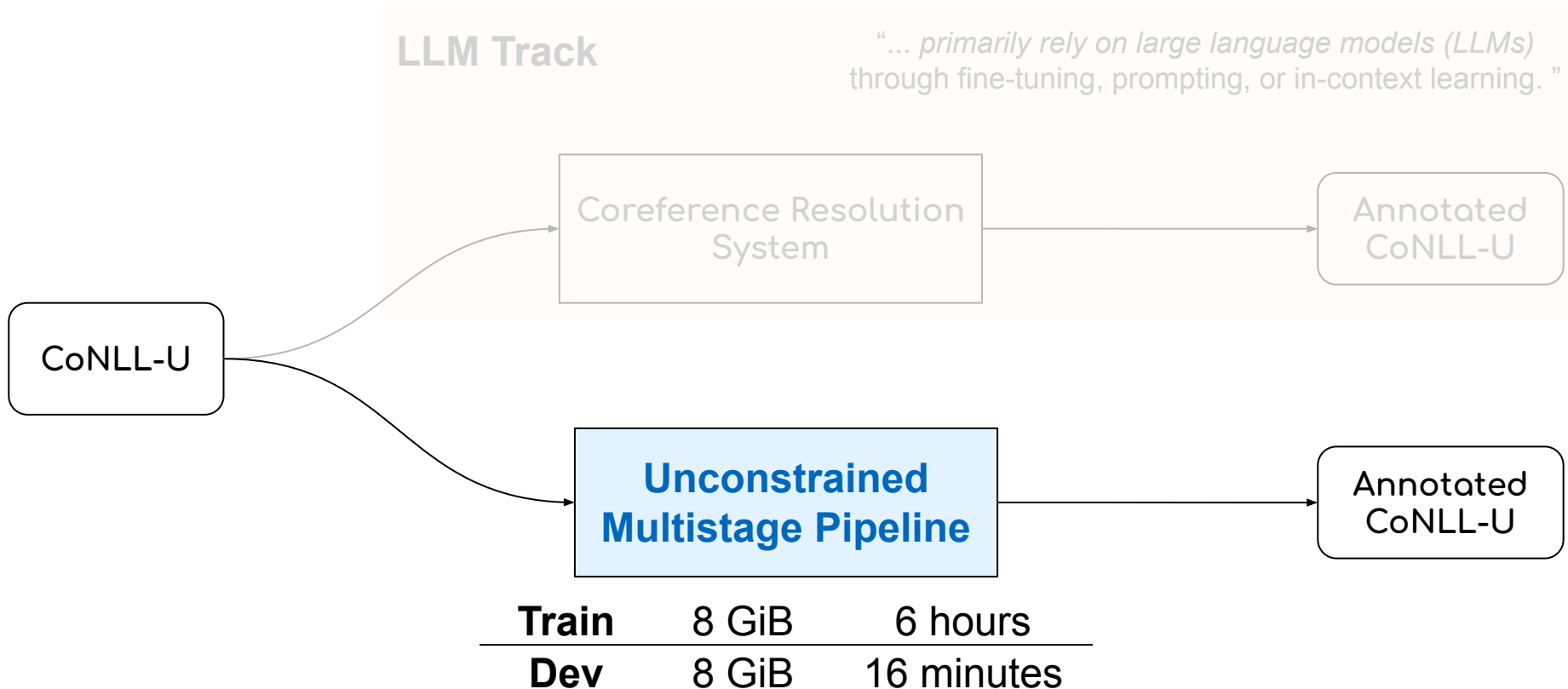
< 3 hours

7.2 GiB

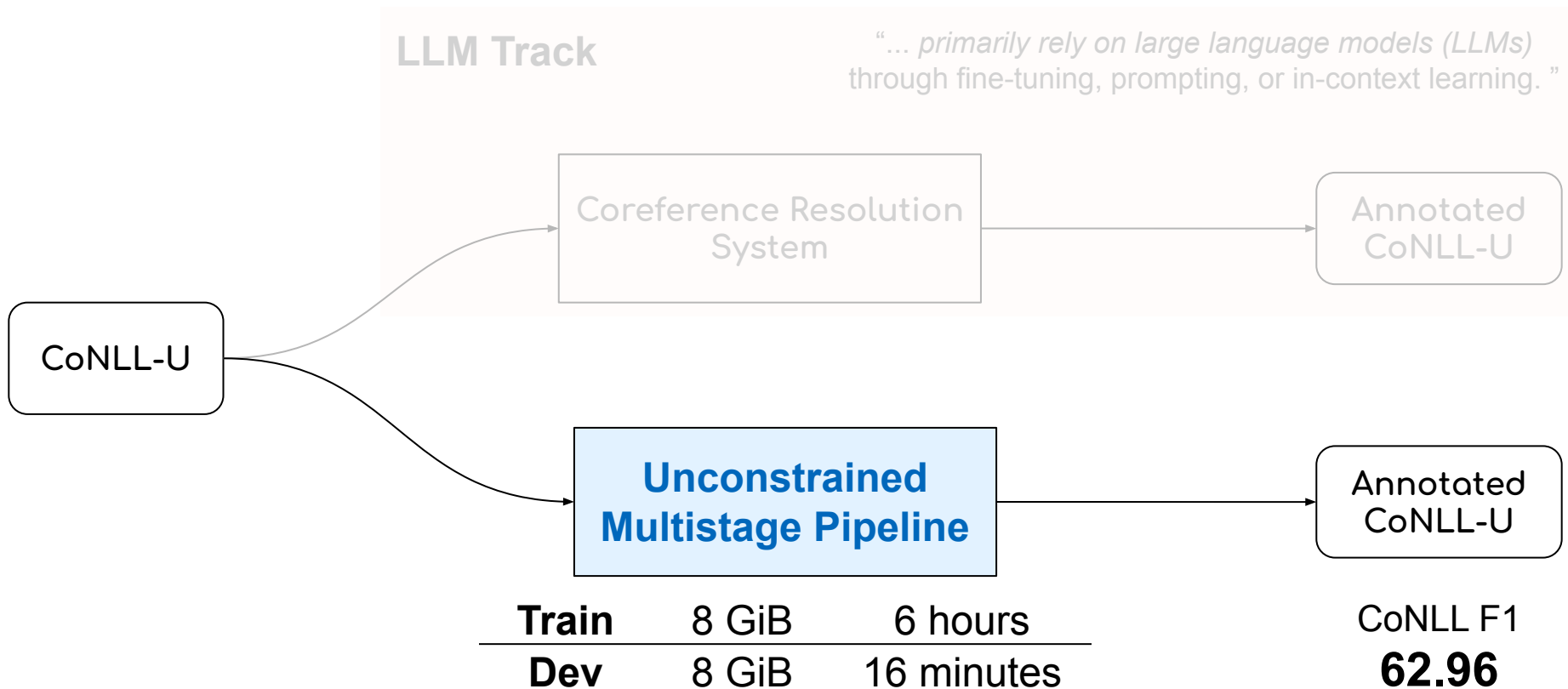
< 5 hours

- **8 GiB**
- **< 1 hour**

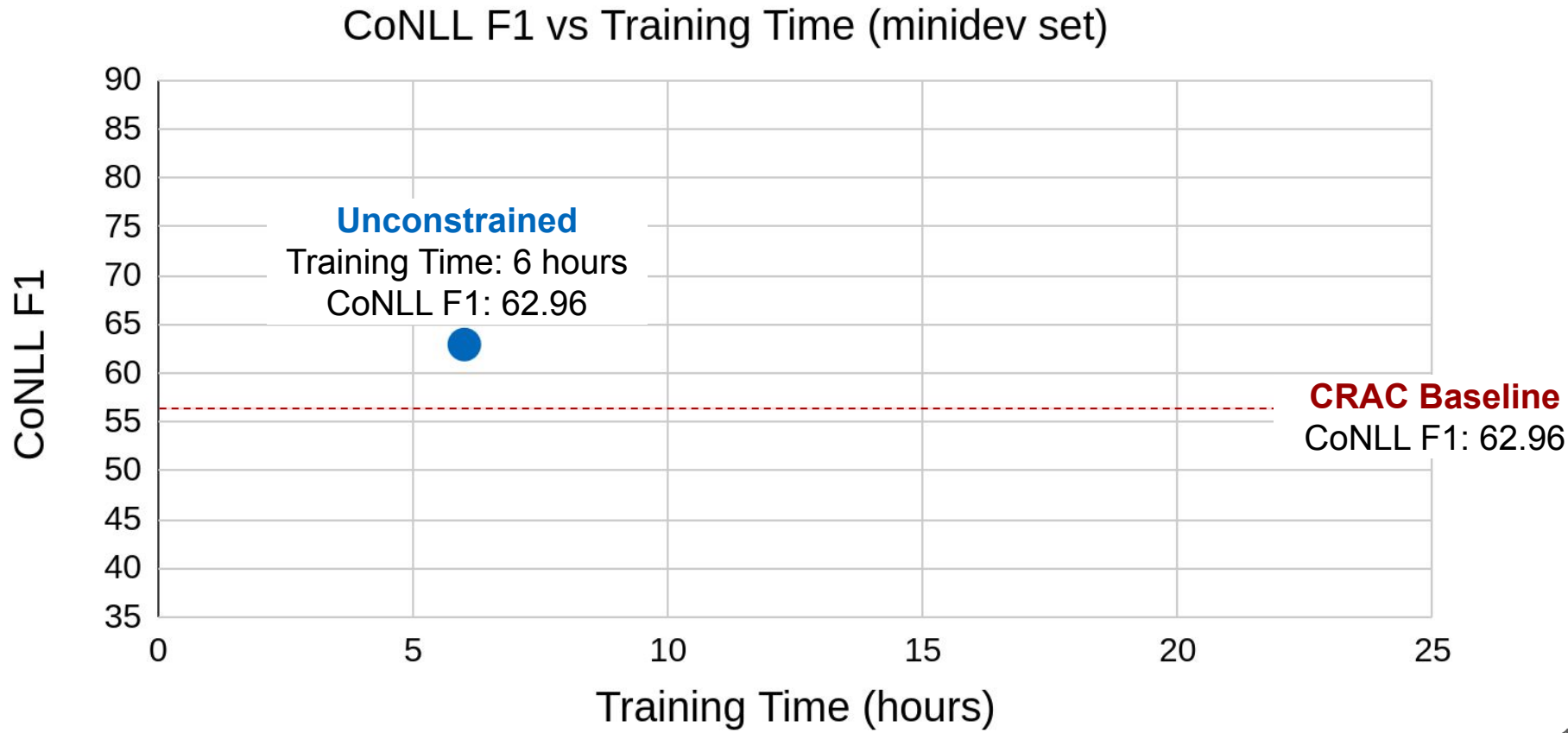
Unconstrained Submission: Training Resources



Unconstrained Submission: Results



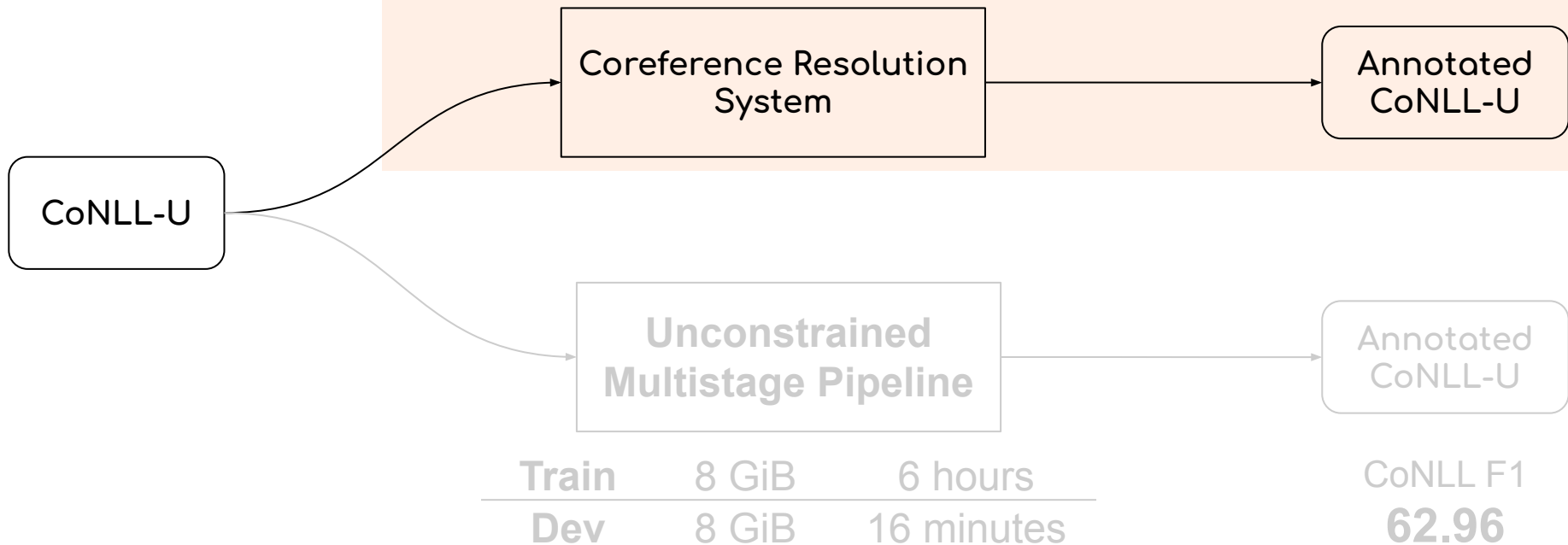
Comparison of Models



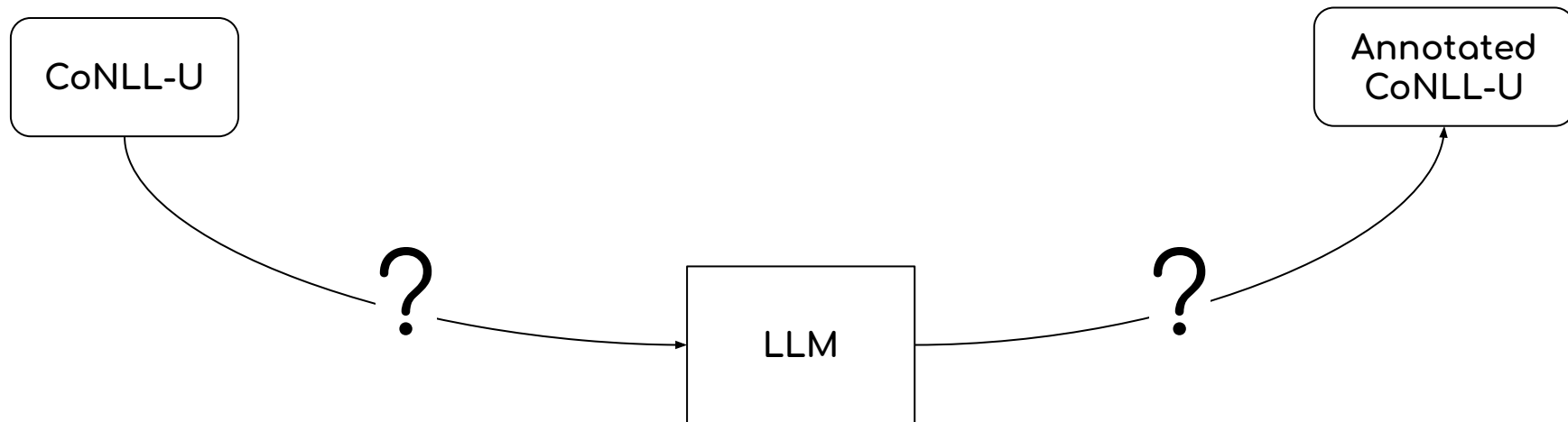
LLM Track

LLM Track

“... primarily rely on large language models (LLMs) through fine-tuning, prompting, or in-context learning.”



LLM Track



LLM Track: Provided Conversion Scripts

```
# newdoc id = 11_alices_adventures_in_wonderland
# global.Entity = eld-etypc-head-other
# newpar
# sent_id = s656
# text = CHAPTER I. Down the Rabbit-Hole Alice was beginning to get very tired of sitting by her sister on the bank, and of having
do: once or twice she had peeped into the book her sister was reading, but it had no pictures or conversations in it, 'and what
book,' thought Alice 'without pictures or conversations?'
1 CHAPTER Chapter PRON NNP Number=Sing 8 nsubj
2 I. I. NUN CO NumForm=NonNum|NumType=Card 1 prep - -
3 Down down ADP IN 5 case - -
4 the the DET DT Definite=Def|Prontype=Art 5 det - Entity=(e6351-place-2
5 Rabbit-hole Rabbit-hole PRON NNP Number=Sing 1 nmod - Entity=(e6351)
6 Alice Alice PRON NNP Number=Sing 8 nsubj - Entity=(e6352-person-1)
7 was be AUX VBD Mood=Ind|Number=Sing|Person=3|Tense=Past|VerbForm=Fin 8 aux - -
8 beginning begin VERB VBG Tense=Pres|VerbForm=Part 8 root - -
9 to to PART TO 10 mark - -
10 get get VERB VB VerbForm=Inf 8 xcomp - -
11 very very ADV RB - 12 admod - -
12 tired tired ADO JJ Degree=Pos 10 xcomp - -
13 of of SCONJ IN 14 mark - -
14 sitting sit VERB VBG Tense=Pres|VerbForm=Part 12 advcl - -
15 by by ADP IN 17 case - -
16 her her PRON PRPS Case=Gen|Poss=Yes|Prontype=Prs 17 nmod:poss - Entity=(e6353-person-2(e
17 sister sister NOUN NN Number=Sing 14 obl - Entity=(e6353)
18 on on ADP IN 20 case - -
19 the the DET DT Definite=Def|Prontype=Art 20 det - Entity=(e6354-place-2
20 bank bank NOUN NN Number=Sing 14 obl - Entity=(e6354)SpaceAfter=No
21 , , PUNCT , 24 punct - -
22 and and CONJ CC - 24 cc - -
23 of of SCONJ IN 24 mark - -
24 having have VERB VBG Tense=Pres|VerbForm=Part 14 con] - -
25 nothing nothing PRON NN Number=Sing|Prontype=Neg 24 obj - -
26 to to PART TO 27 mark - -
27 do do VERB VB VerbForm=Inf 25 acl - SpaceAfter=No
28 : : PUNCT : 34 punct - -
29 once once ADV RB NumForm=Word|NumType=Multi 34 advmod - -
30 or or CONJ CC - 31 cc - -
31 twice twice ADV RB NumForm=Word|NumType=Multi 29 con] - -
32 she she PRON PRP Case=Nom|Gender=Fem|Number=Sing|Person=3|Prontype=Prs 34 nsubj - Entity=(
33 had have AUX VBD Mood=Ind|Number=Sing|Person=3|Tense=Past|VerbForm=Fin 34 aux - -
34 peeped peep VERB VBN Tense=Past|VerbForm=Part 8 parataxis - -
35 into into ADP IN 37 case - -
36 the the DET DT Definite=Def|Prontype=Art 37 det - -
37 book book NOUN NN Number=Sing 14 obl - -
38 her her PRON PRPS Case=Gen|Gender=Fem|Number=Sing|Person=3|Poss=Yes|Prontype=Prs 39 nmod:poss
```

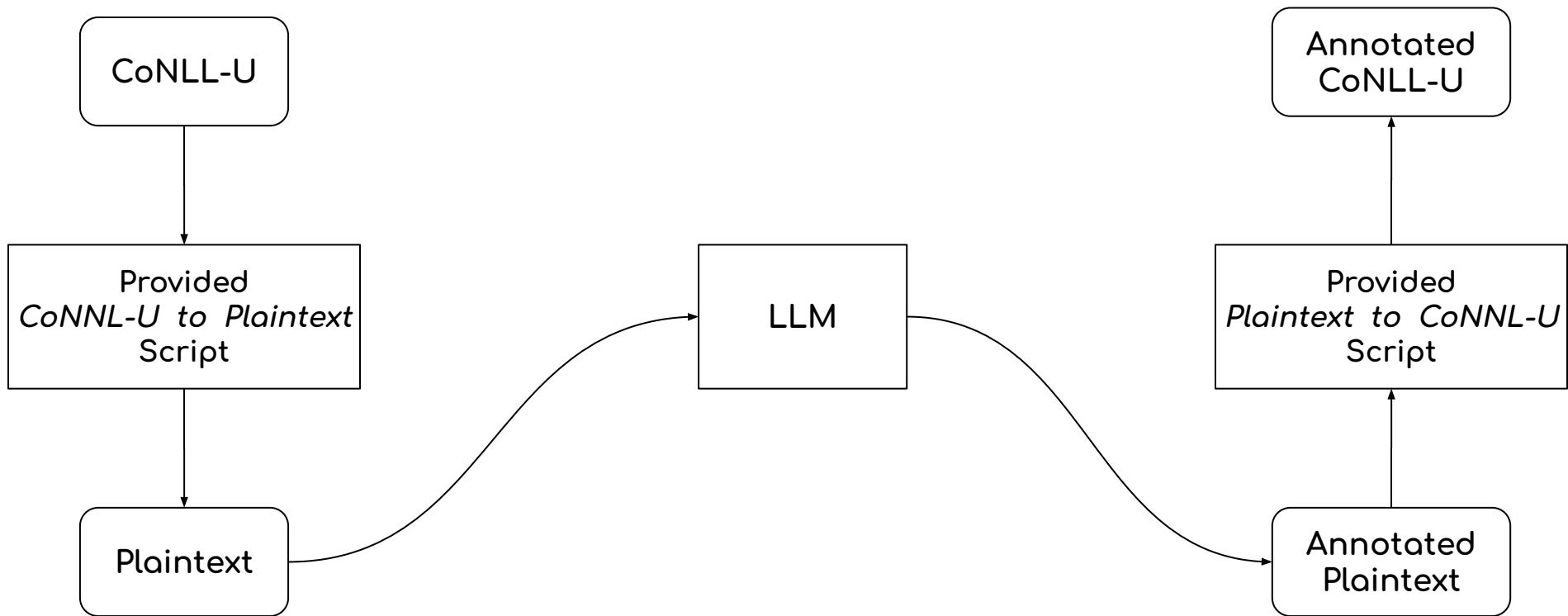
LLM

Annotated
CoNLL-U

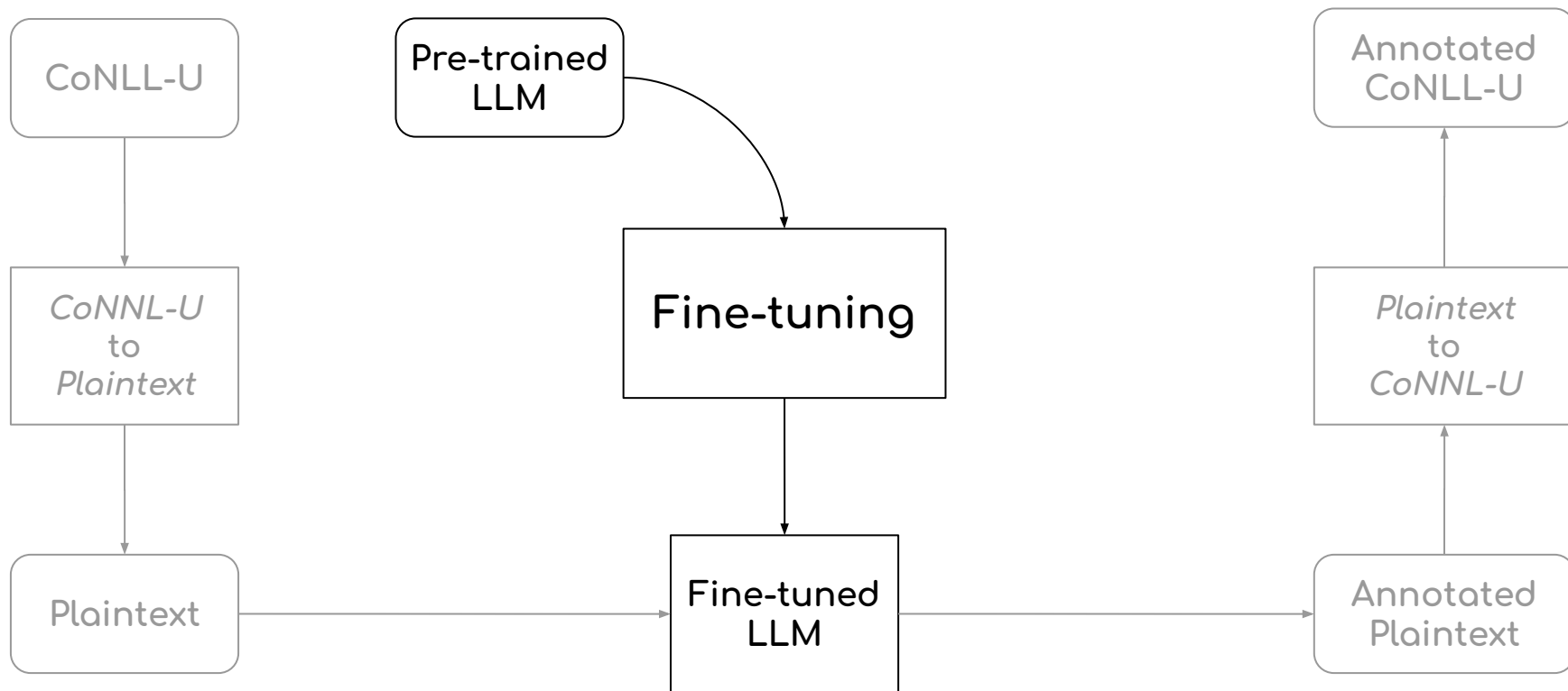
?

LLMs are not designed to process CoNLLU-formatted data

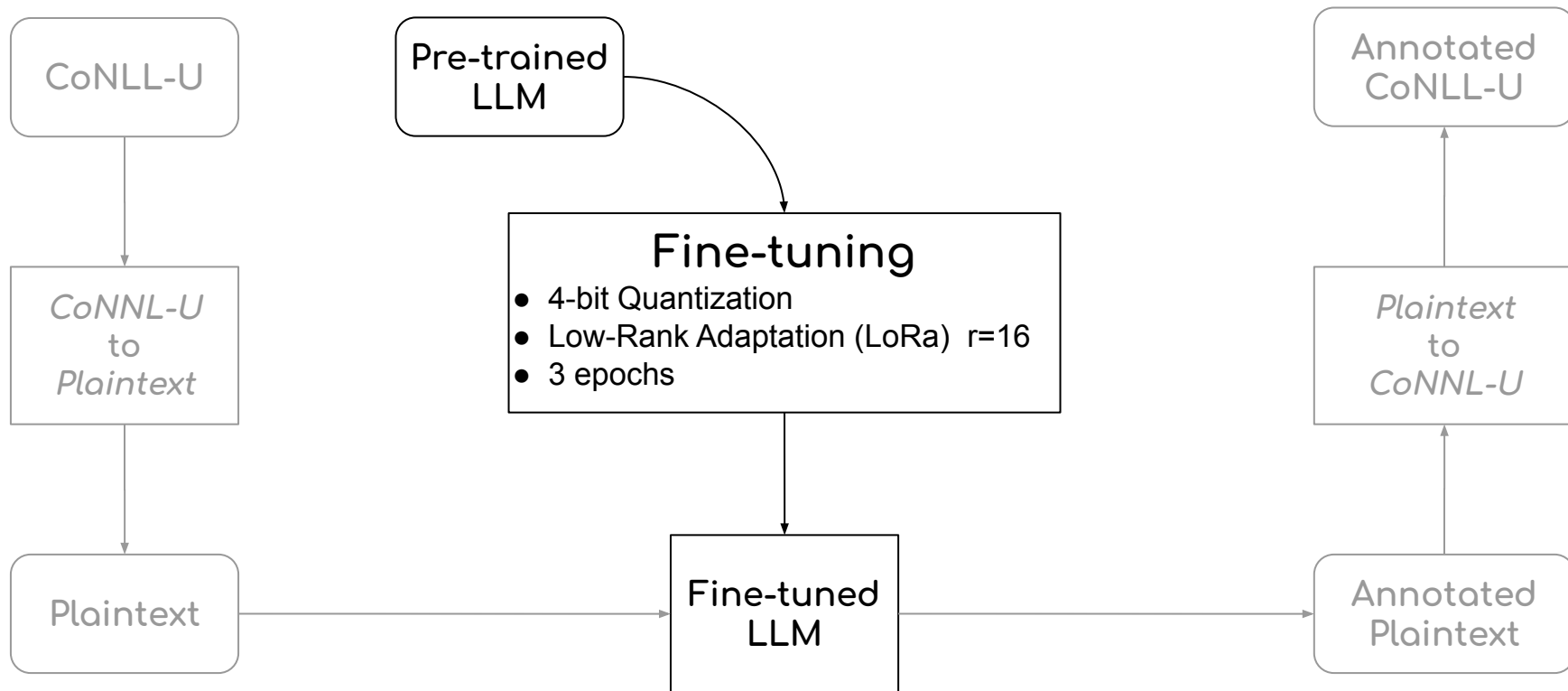
LLM Track: Provided Conversion Scripts



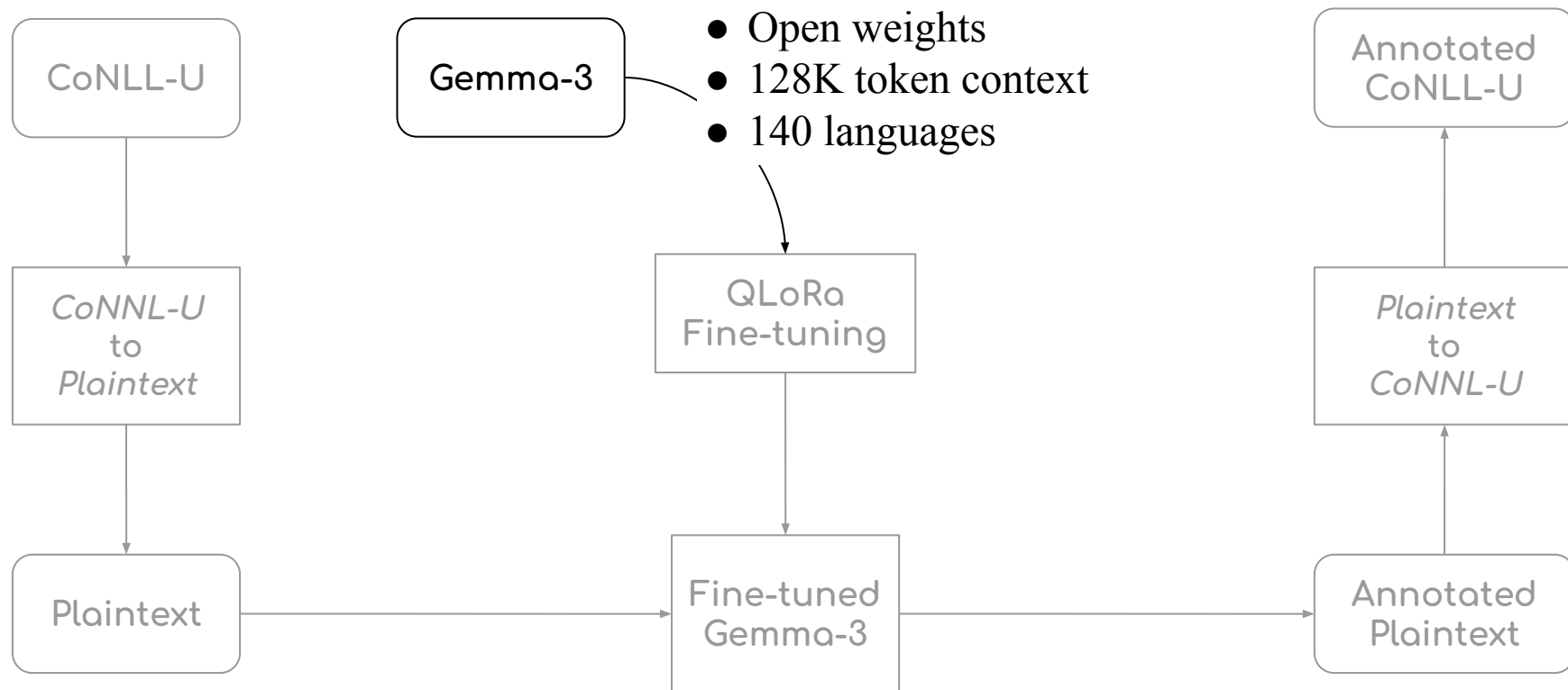
LLM Fine-tuning



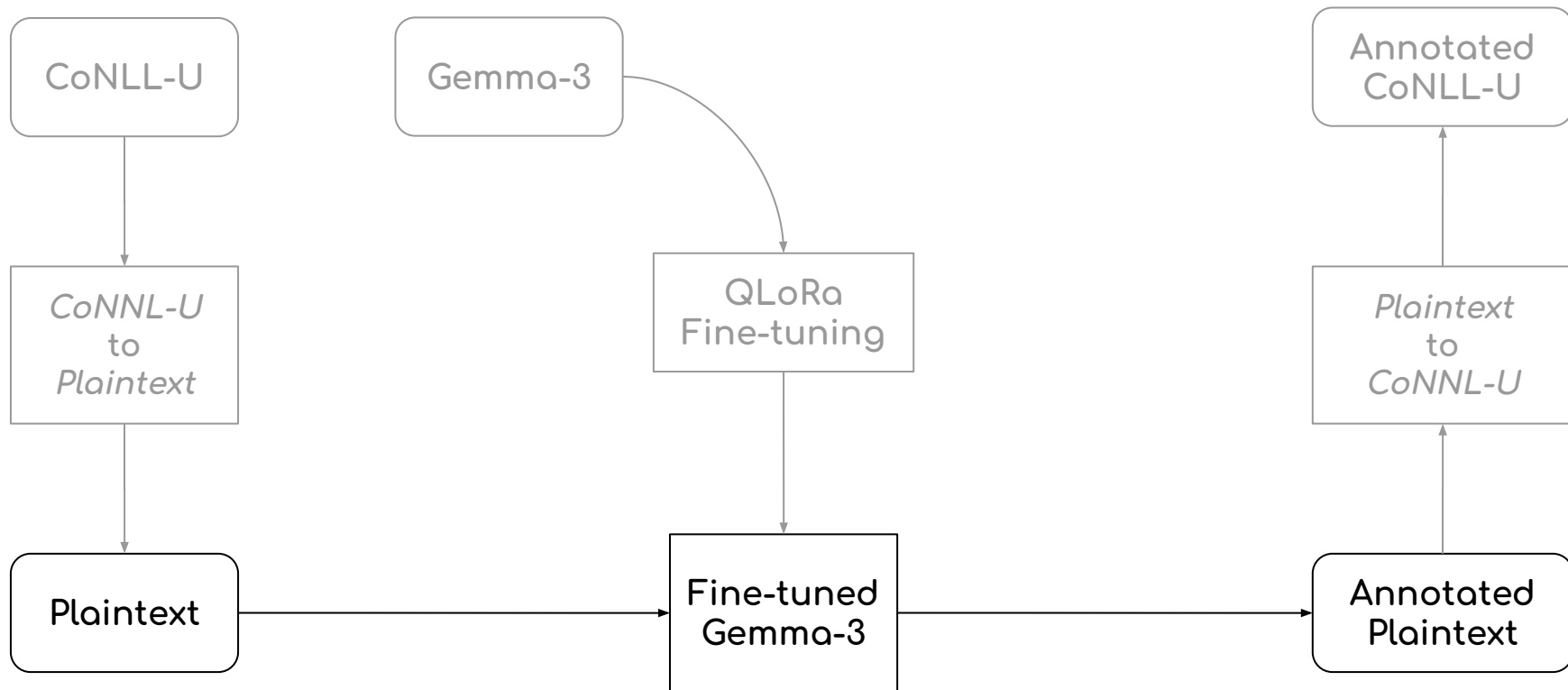
LLM Fine-tuning



Pre-trained LLM Choice: Gemma-3 instruction-tuned (IT)



Input Formatting



Gemma-3 Prompt Template

SYSTEM INSTRUCTION

TEXT INPUT

EXPECTED MODEL OUTPUT

```
<start_of_turn>user
```

```
You are a linguist, expert in anaphora and coreference resolution.  
Annotate in the input sentences which nouns, pronouns and other expressions  
refer to the same entity.  
Do only insert annotations. Do not insert extra linguistic material, nor  
punctuation markers and do not delete elements from the input texts.
```

```
Input: *PLAINTEXT*
```

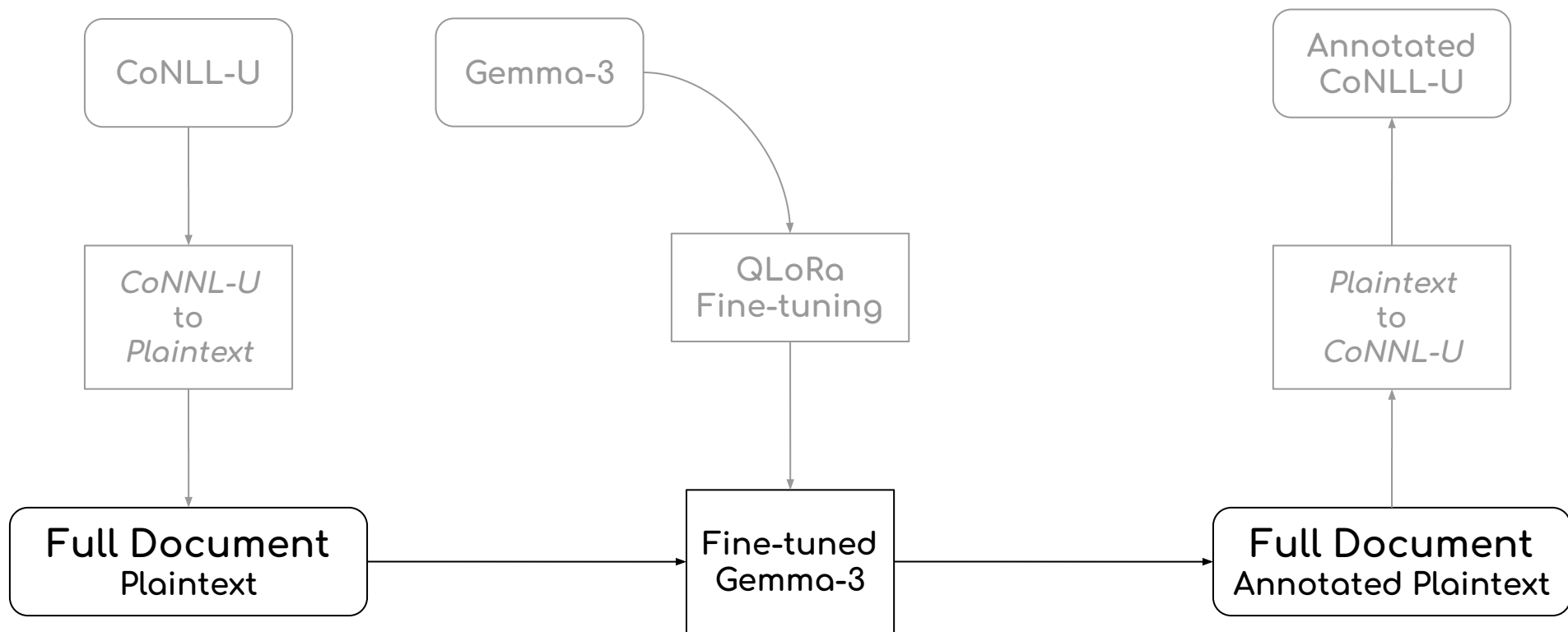
```
<end_of_turn>
```

```
<start_of_turn>model
```

```
*COREFERENCE ANNOTATED PLAINTEXT*
```

```
<end_of_turn>
```

Full Document Annotation



Gemma-3: Prompt Template

SYSTEM INSTRUCTION

TEXT INPUT

EXPECTED MODEL OUTPUT

```
<start_of_turn>user
```

```
You are a linguist, expert in anaphora and coreference resolution.  
Annotate in the input sentences which nouns, pronouns and other expressions  
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punctuation markers and do not delete elements from the input texts.
```

```
Input: *PLAINTEXT*
```

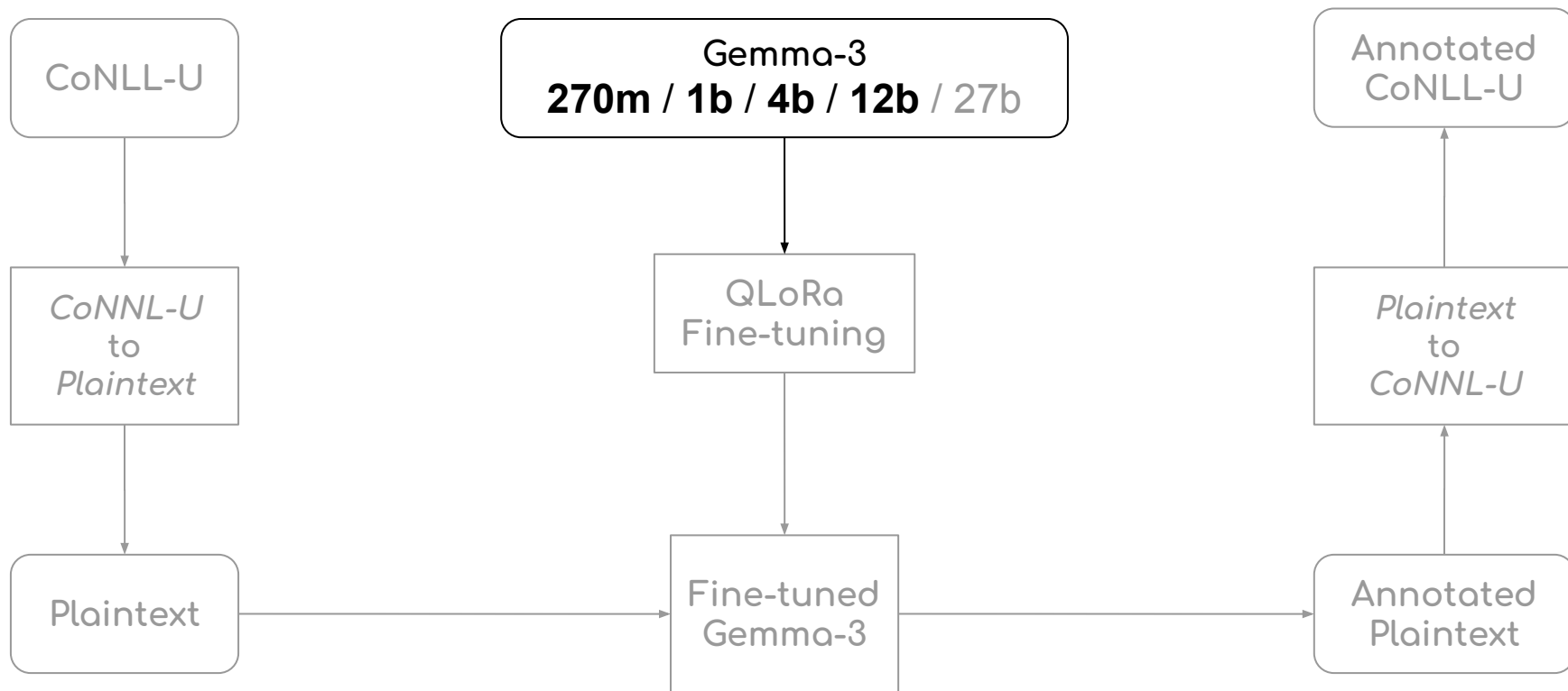
```
<end_of_turn>
```

```
<start_of_turn>model
```

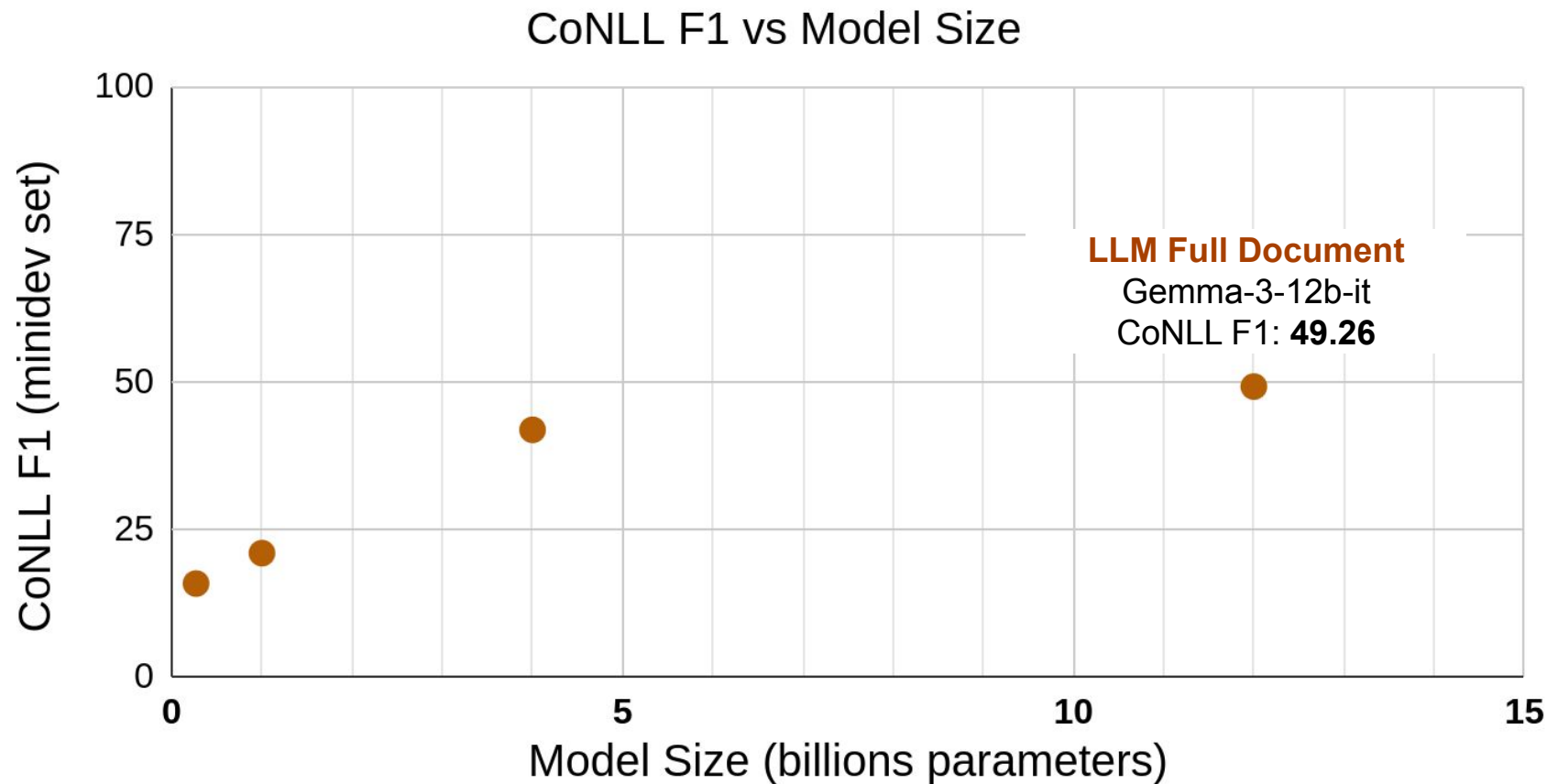
```
*COREFERENCE ANNOTATED SENTENCE BATCH*
```

```
<end_of_turn>
```

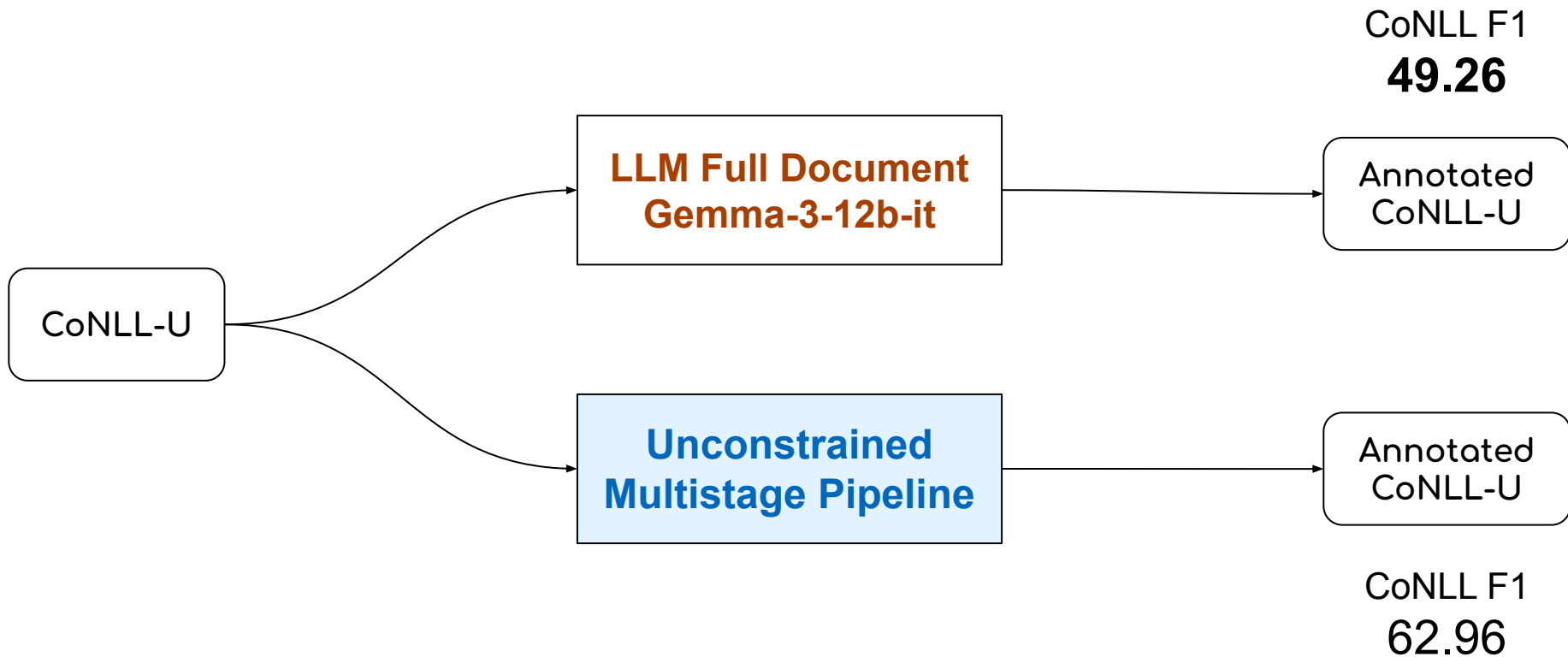
Full Document Annotation: Model Size Impact



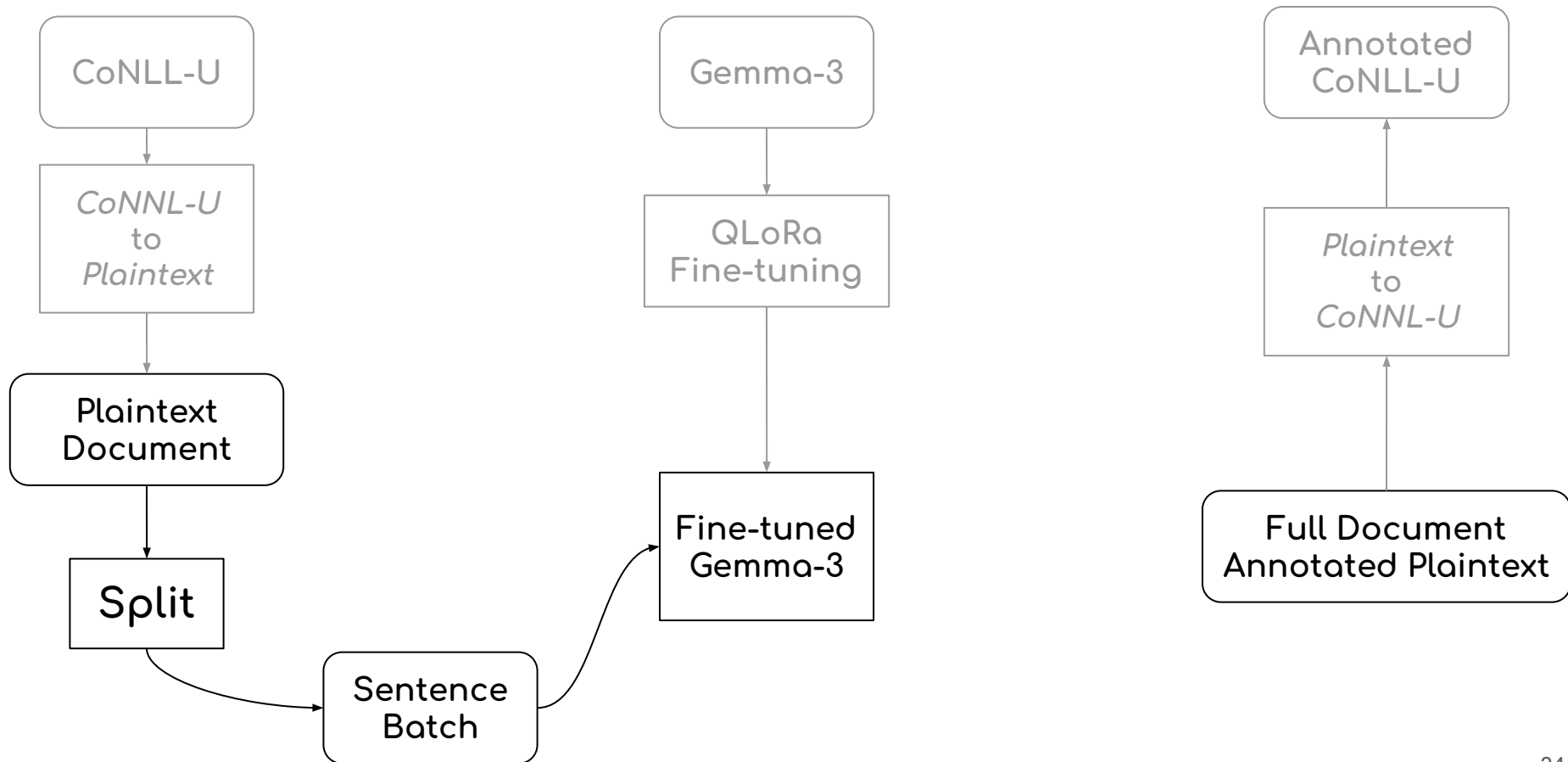
Full Document Annotation: Model Size Impact



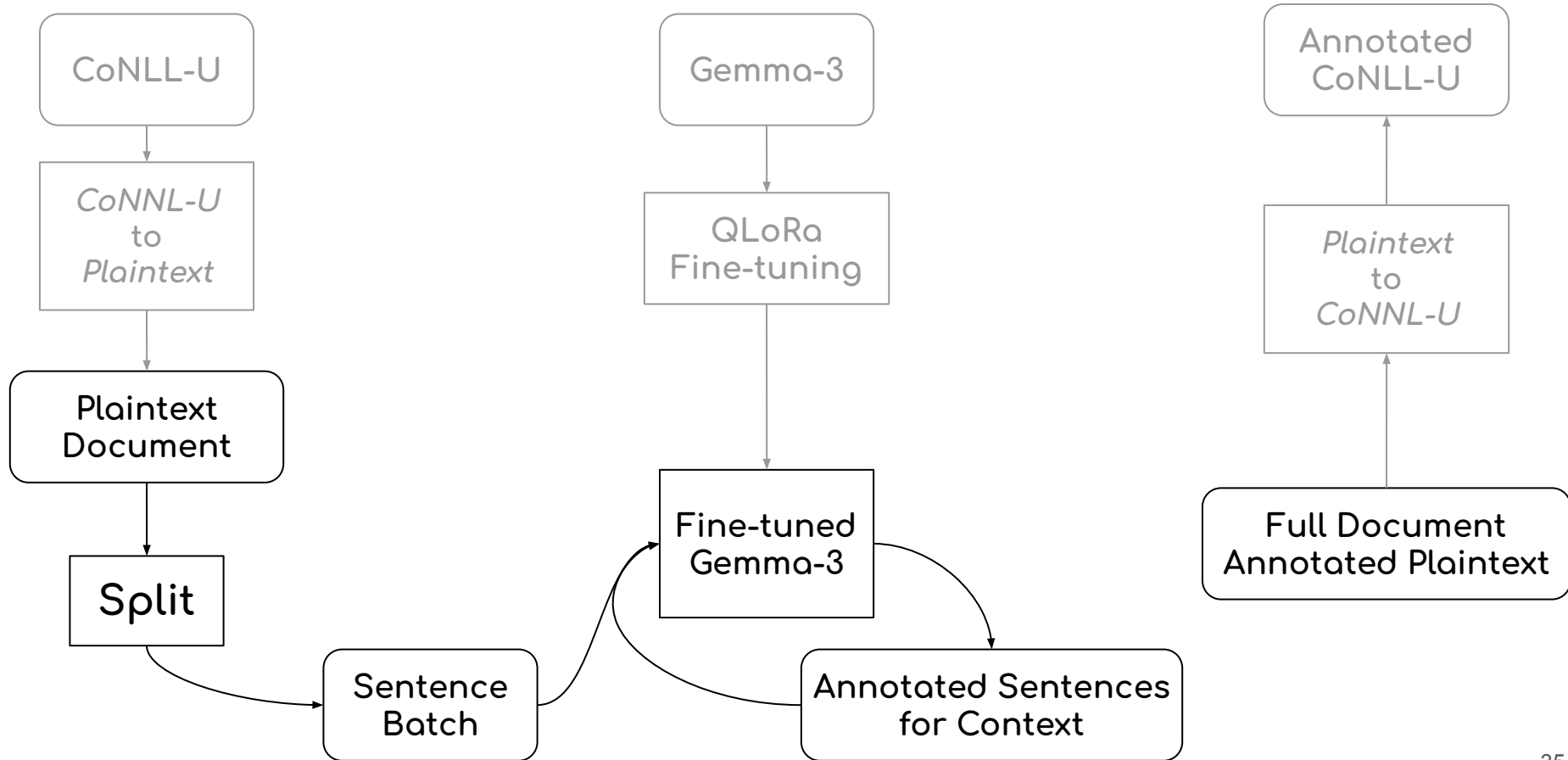
Unconstrained > LLM Full Document



Incremental Sentence Batch Annotation



Incremental Sentence Batch Annotation



Incremental Sentence Batch Annotation: Prompt Template

SYSTEM INSTRUCTION

PREVIOUS CONTEXT

TEXT INPUT

EXPECTED MODEL OUTPUT

<start_of_turn>user

You are a linguist, expert in anaphora and coreference resolution.

Based on the previous context, annotate in the input sentences which nouns, pronouns and other expressions refer to the same entity. Do only insert annotations. Do not insert extra linguistic material, nor punctuation markers and do not delete elements from the input texts.

Previous context: *ANNOTATED SENTENCES FROM PREVIOUS BATCHES*

Input: *PLAINTEXT SENTENCE BATCH*

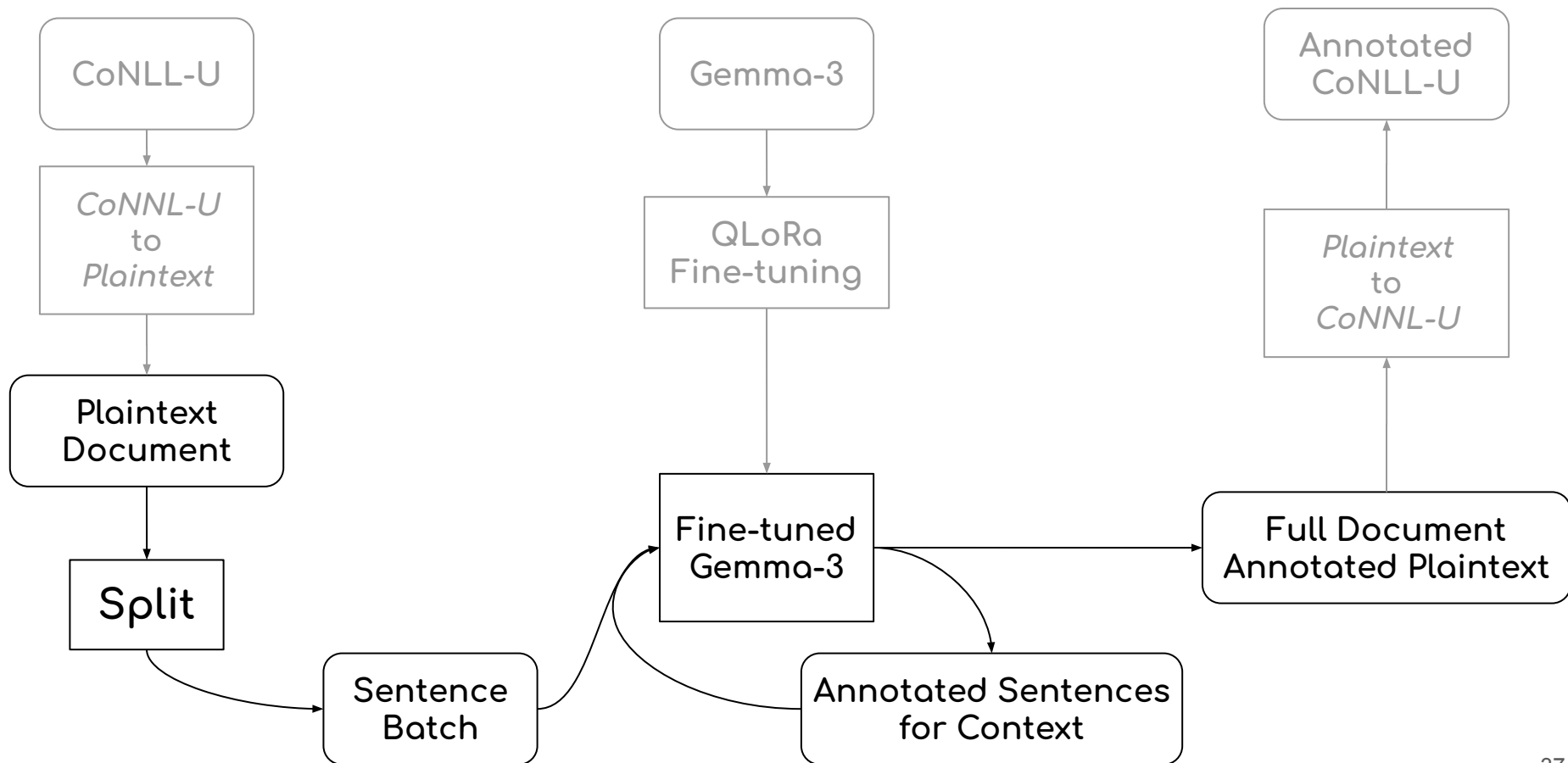
<end_of_turn>

<start_of_turn>model

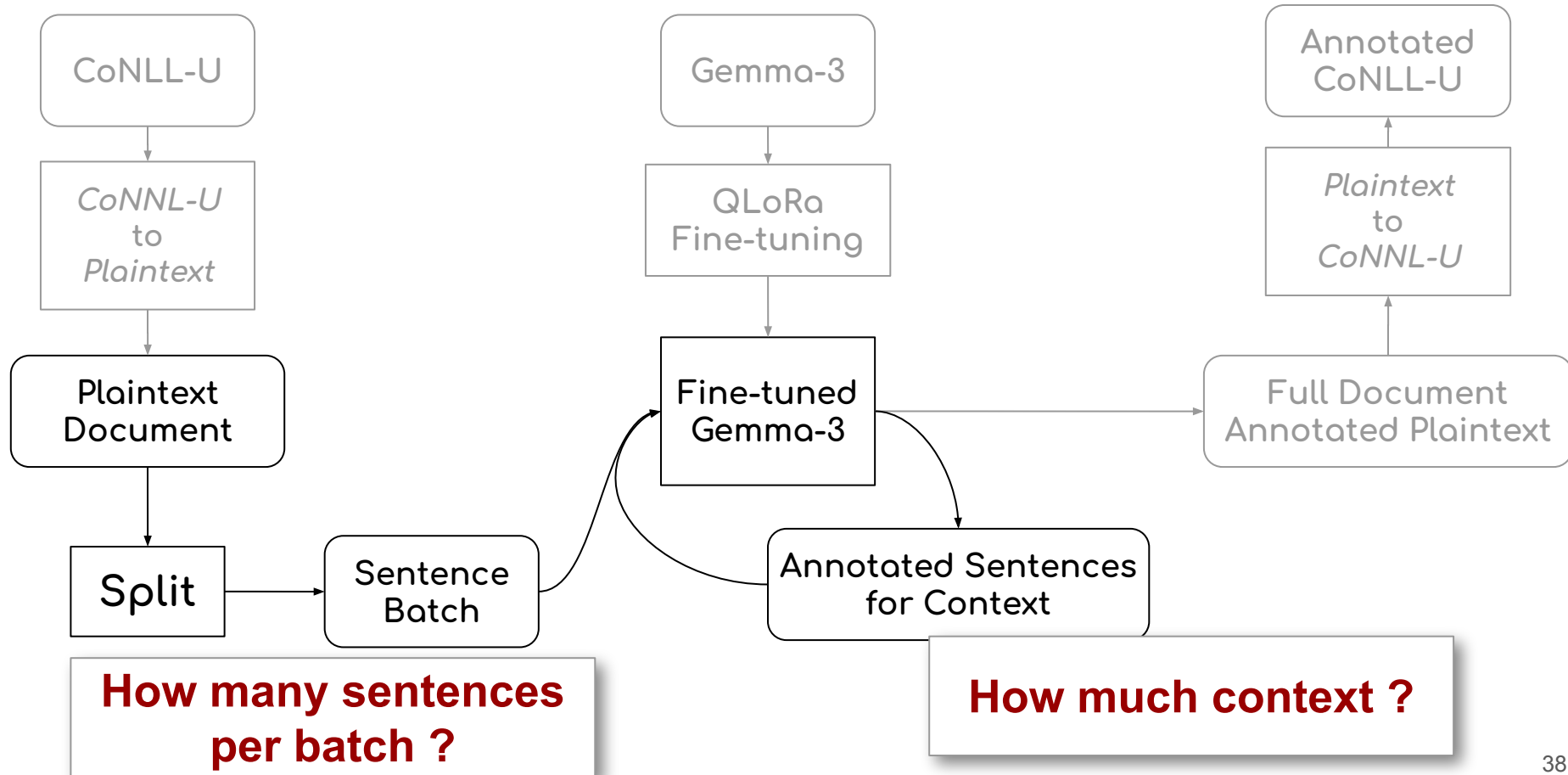
COREFERENCE ANNOTATED SENTENCE BATCH

<end_of_turn>

Incremental Sentence Batch Annotation



Incremental Sentence Batch Annotation



Annotation Strategies

**How much
context ?**

Previous context (words)



Length of text to annotate (sentences)

LLM Full Document

All sentences

No context

**How many sentences
per batch ?**

Annotation Strategies

How much
context ?

Incremental Sentence-by-Sentence

1 sentence per batch
Maximum available context

Previous context (words)

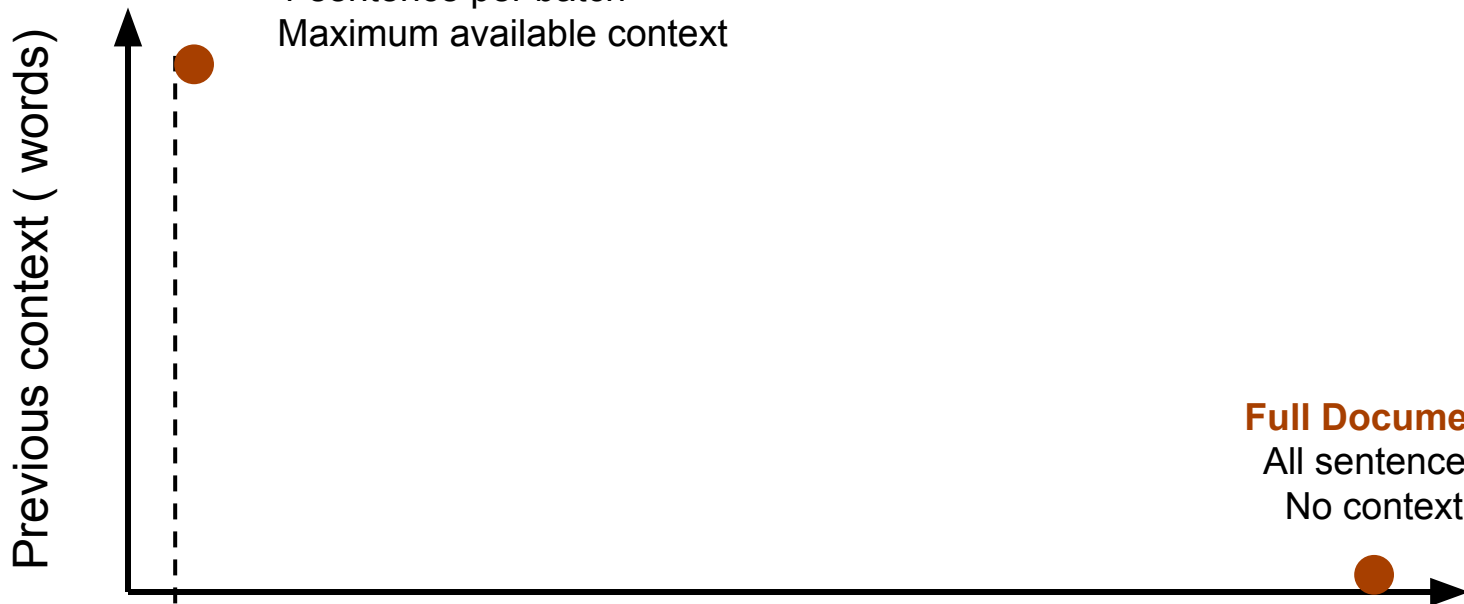
1

Length of text to annotate (sentences)

Full Document

All sentences
No context

How many sentences
per batch ?



Space of Annotation Strategies

How much
context ?

Incremental Sentence-by-Sentence

1 sentence per batch
Maximum available context

Previous context (words)

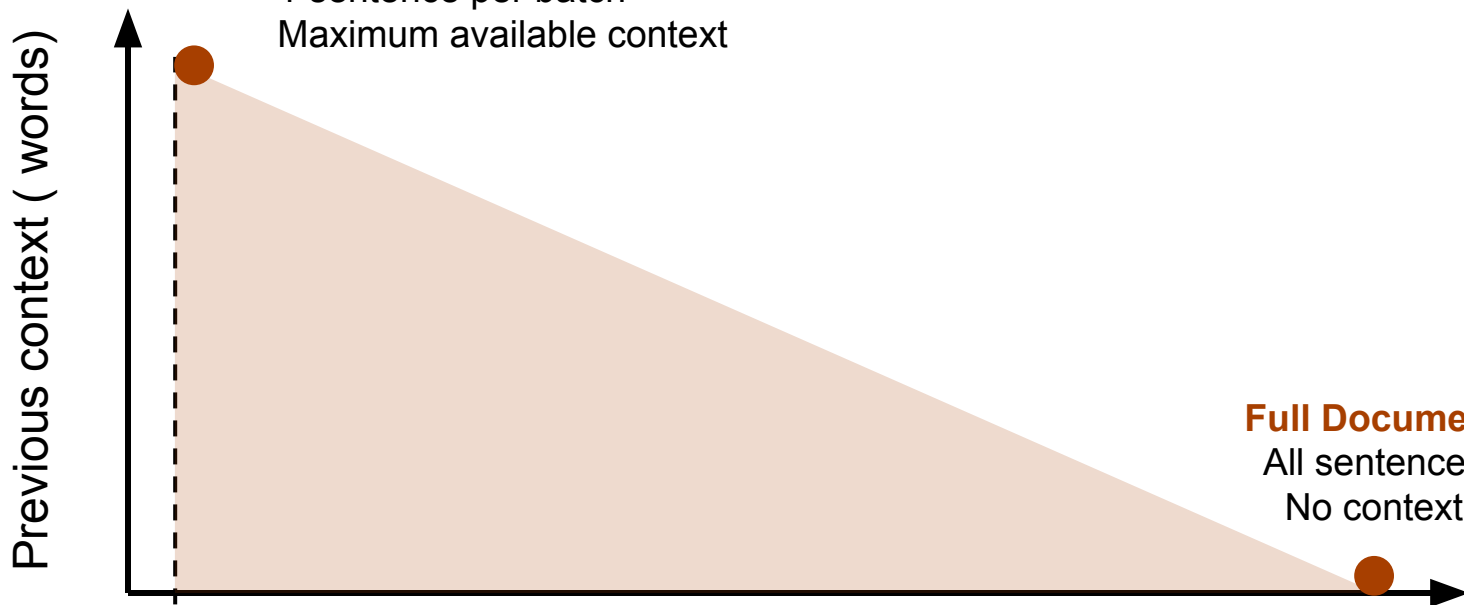
1

Length of text to annotate (sentences)

Full Document

All sentences
No context

How many sentences
per batch ?



Space of Annotation Strategies

How much
context ?

Incremental Sentence-by-Sentence

1 sentence per batch
Maximum available context

Previous context (words)

How does context length
impact coreference
annotation accuracy?

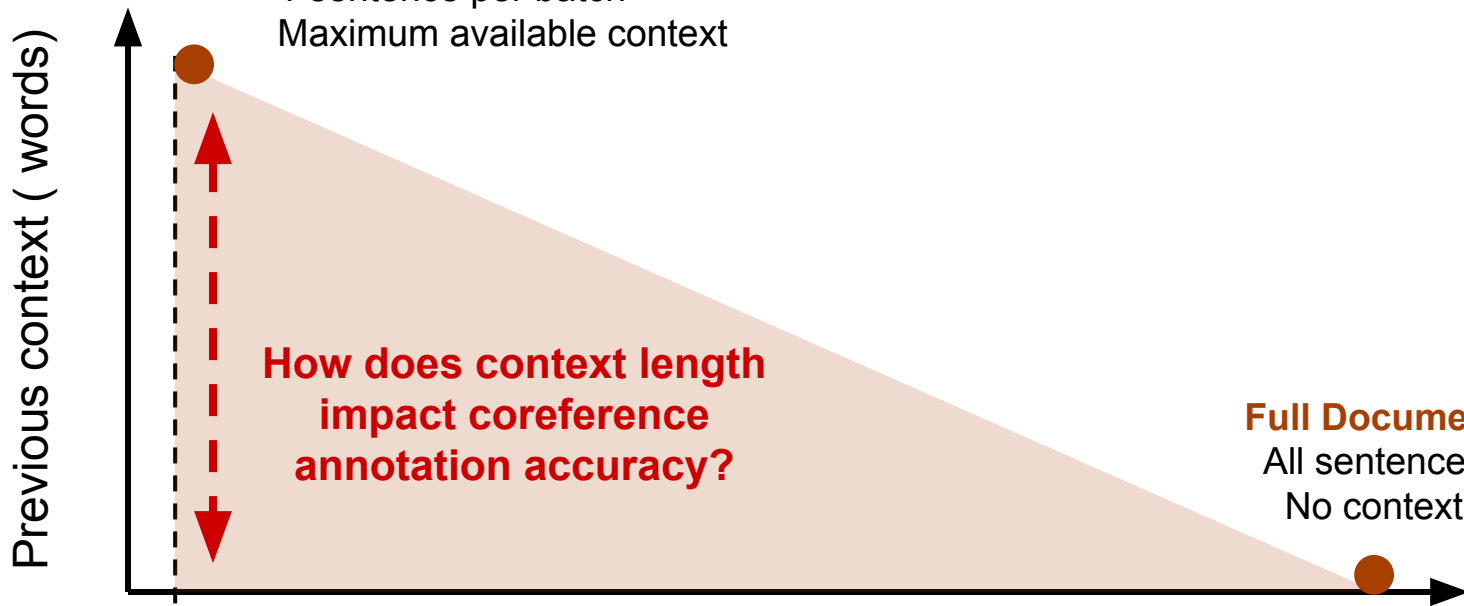
Full Document

All sentences
No context

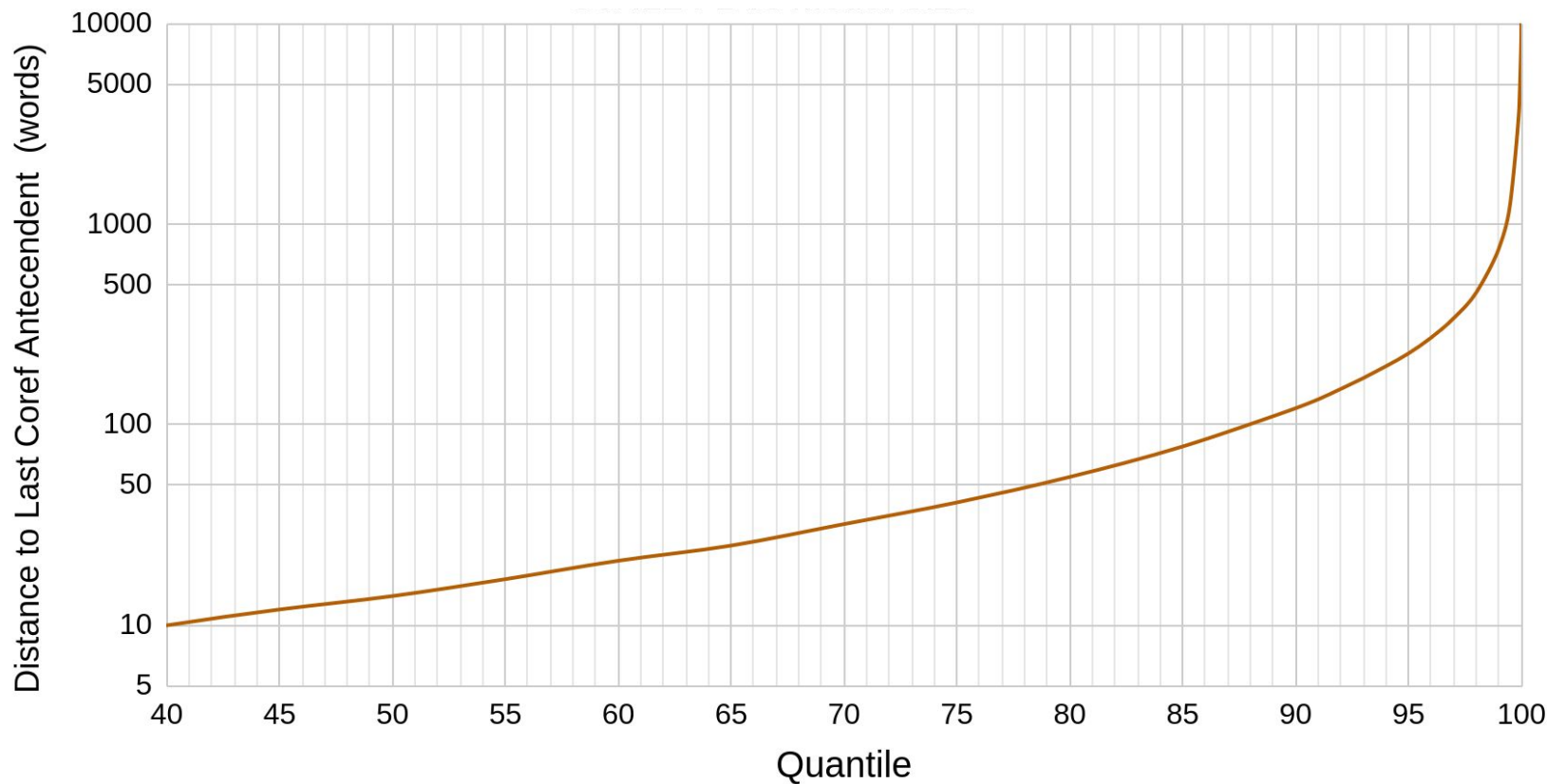
1

Length of text to annotate (sentences)

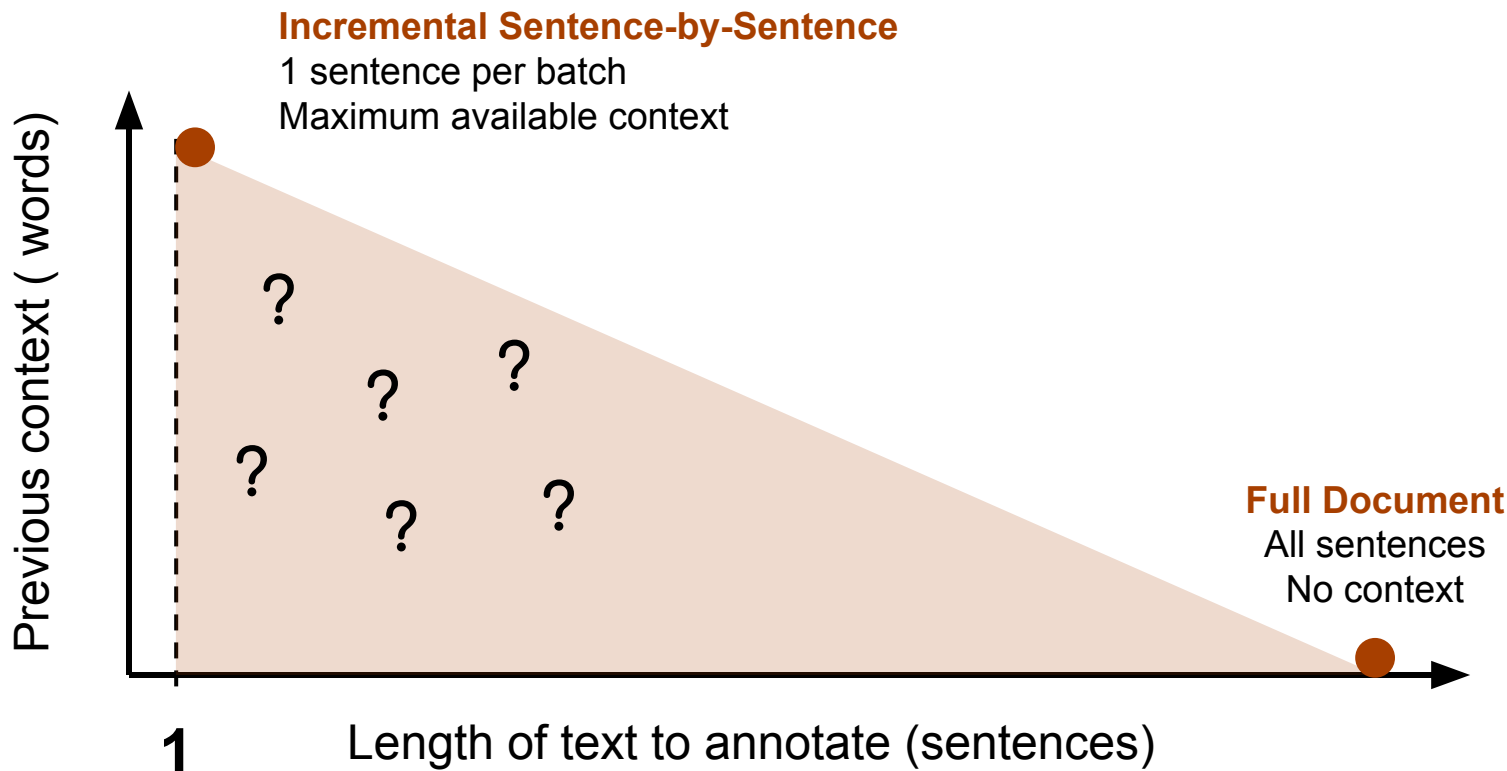
How many sentences
per batch ?



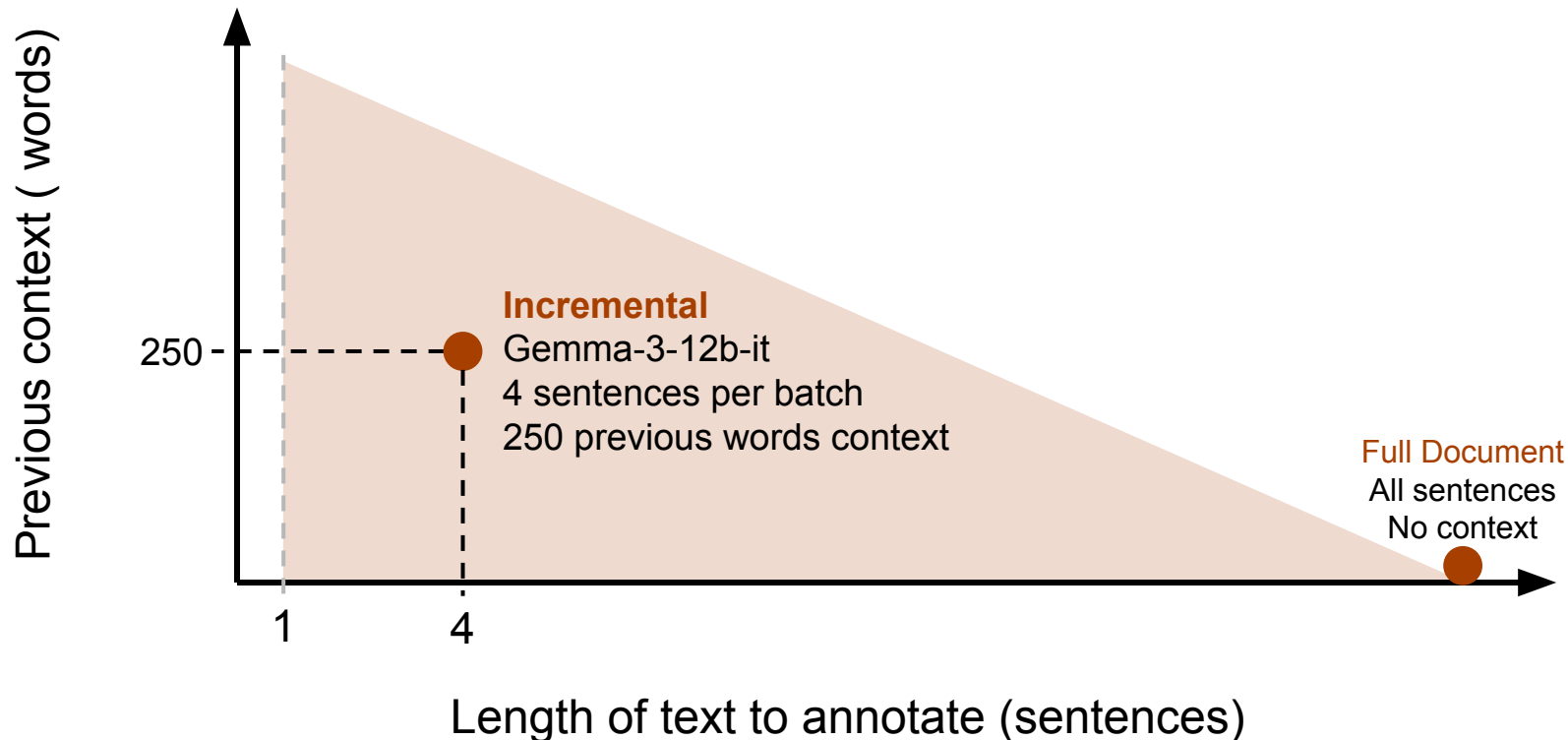
Distance to Last Coreferential Antecedent (train set)



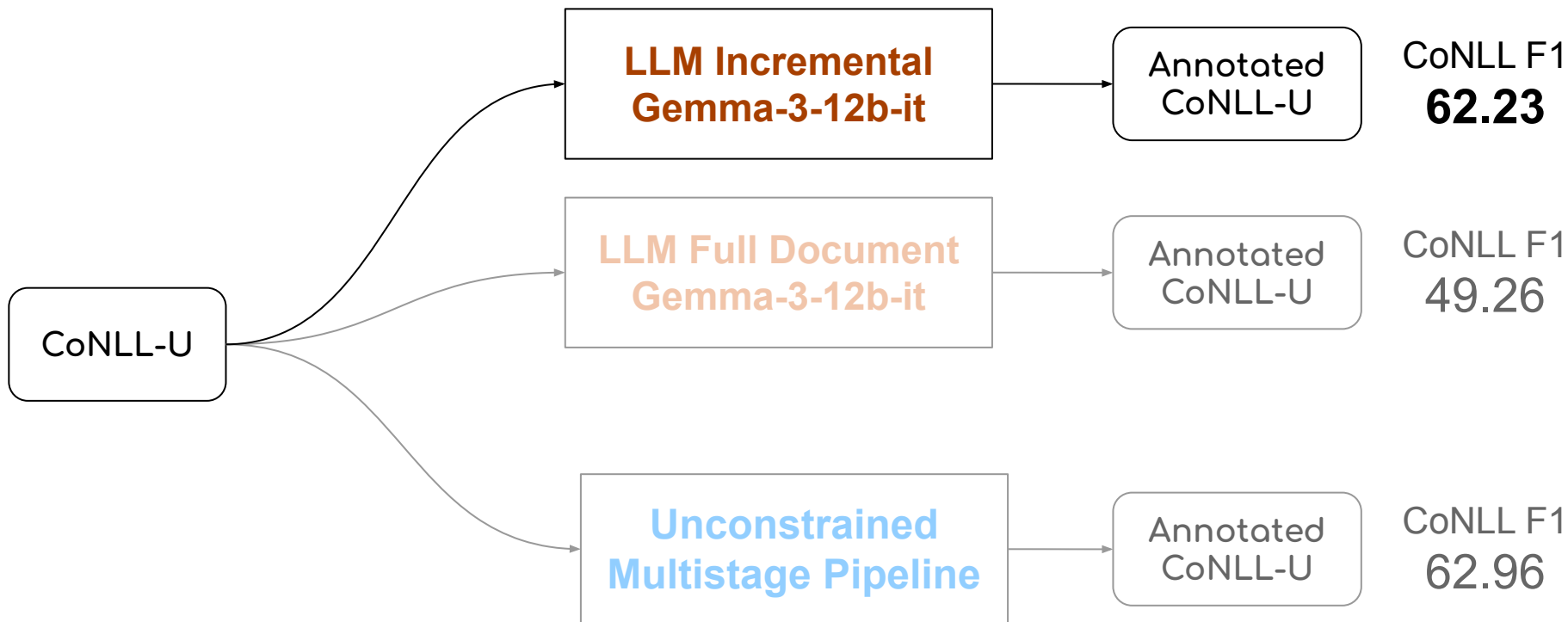
Space of Annotation Strategies: Exploratory Experiments



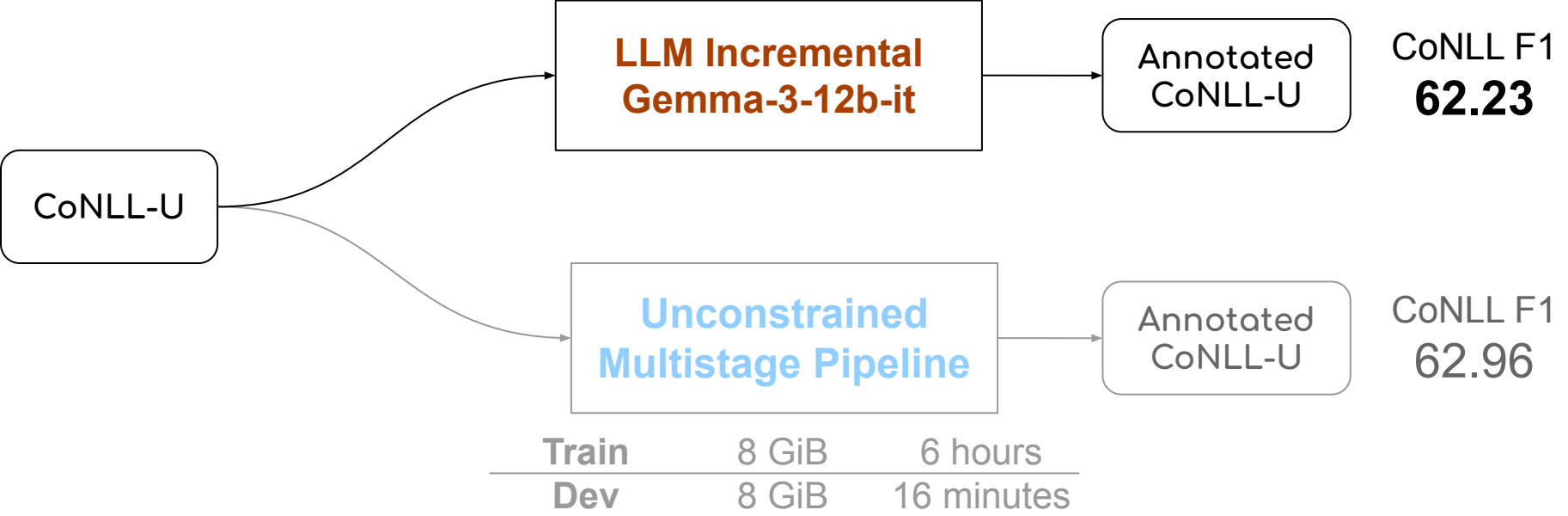
Space of Annotation Strategies



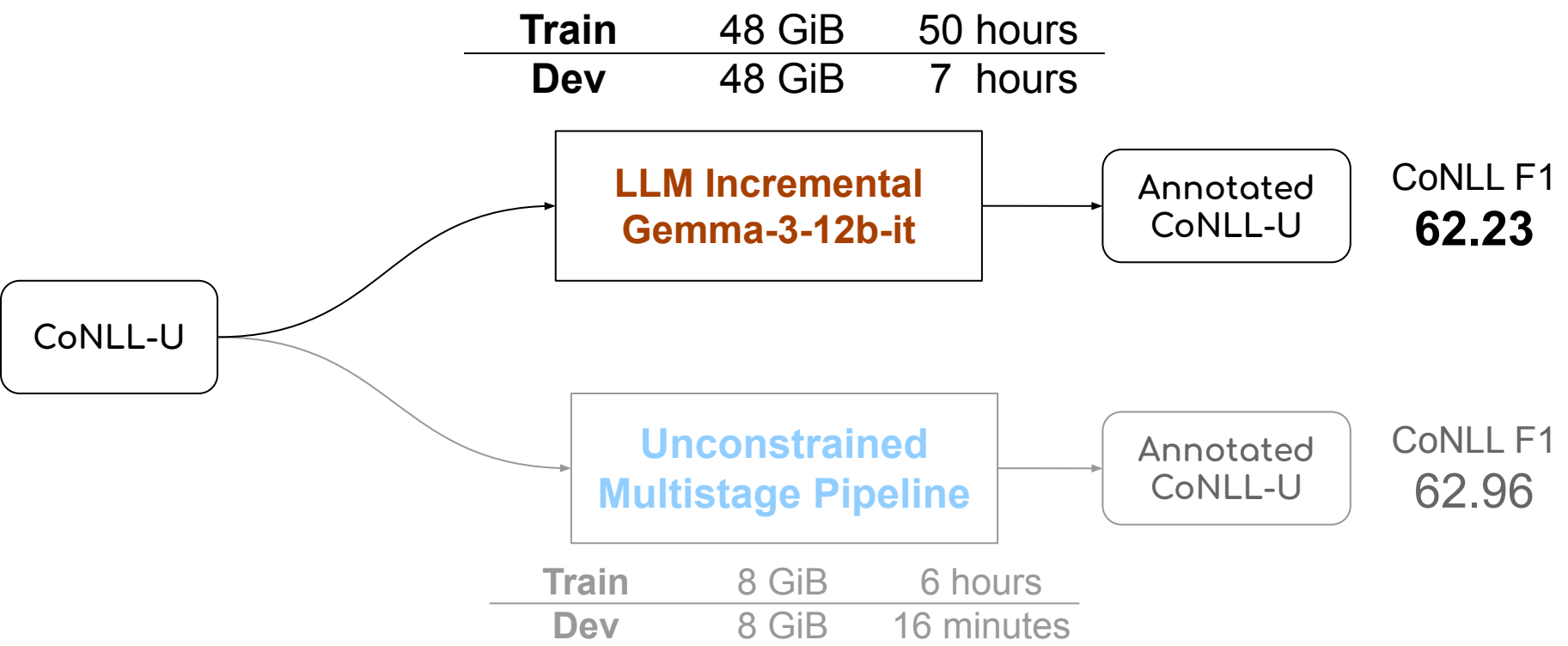
LLM Incremental: Results



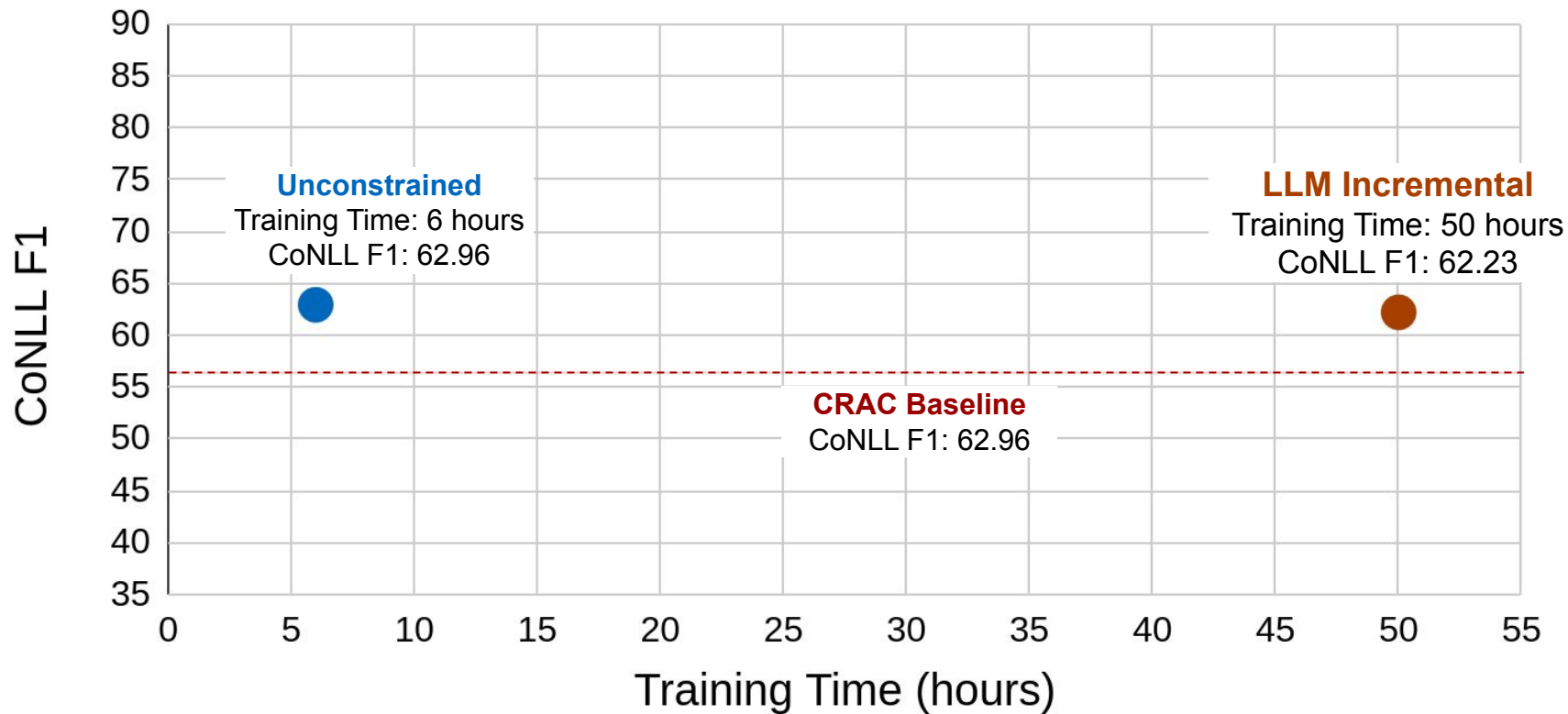
LLM Incremental: Best Model



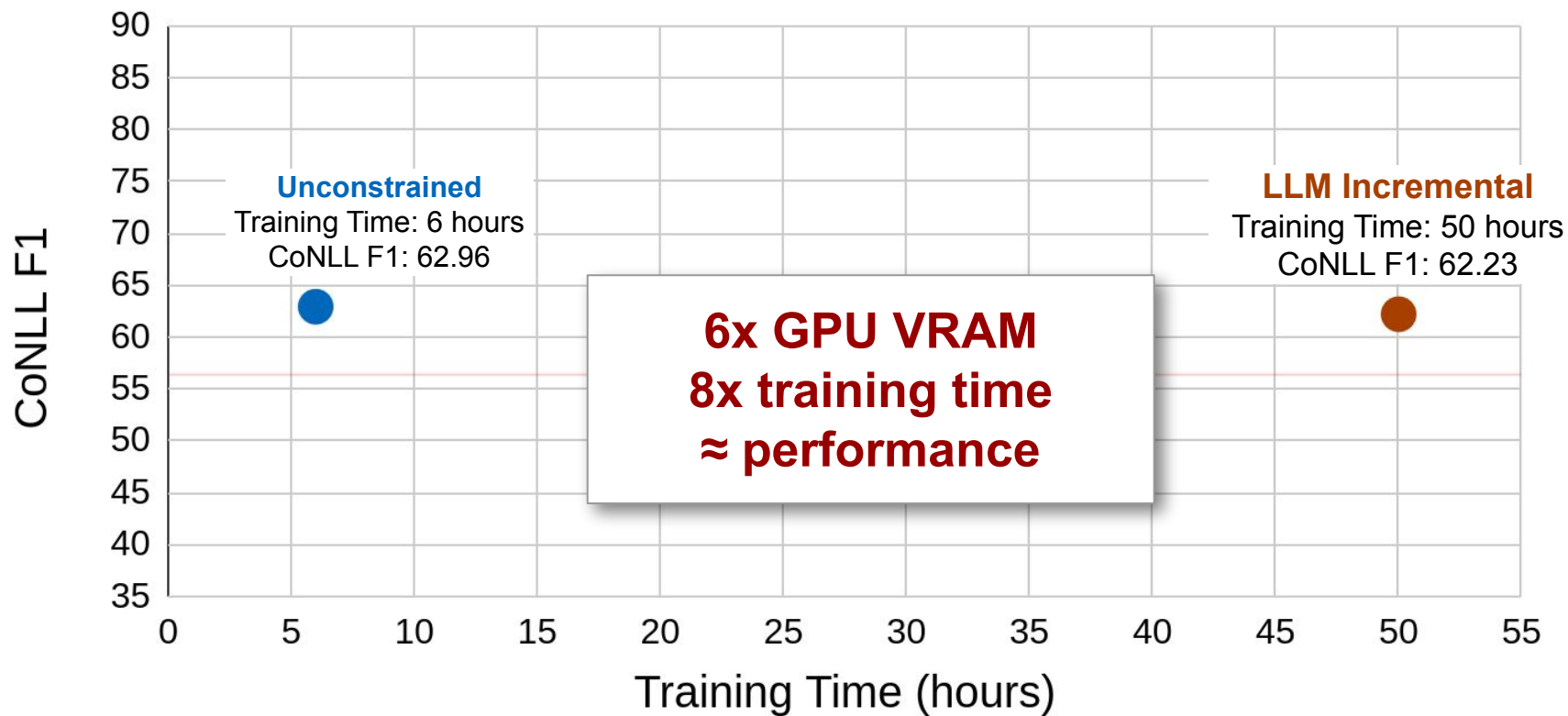
LLM Incremental: Ressources



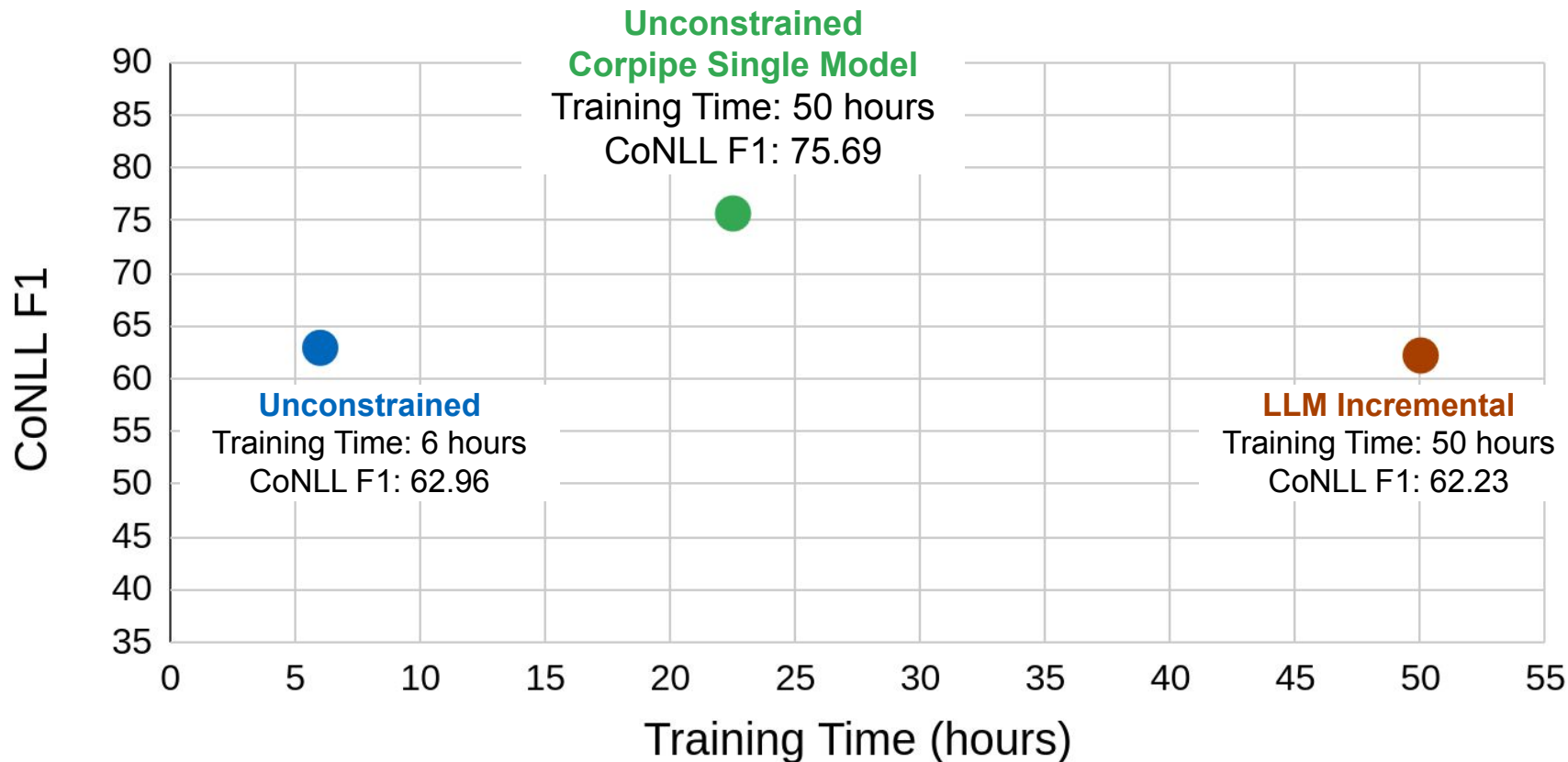
Comparison of Models



Comparison of Models



Comparison of Models: Other Participants



Comparison of Models: Other Participants

system	head-match	
LLM-GLaRef-CRAC25	62.96	←
LLM-NUST-FewShot	61.74	
LLM-PUXCRAC2025	60.09	
LLM-UWB	59.84	
CorPipeEnsemble	75.84	
CorPipeBestDev	75.06	
CorPipeSingle	74.75	←
Stanza	67.81	
GLaRef-Propp	61.57	←
BASELINE-GZ	58.18	
BASELINE	56.01	←

Findings of the Fourth Shared Task on Multilingual Coreference Resolution: Can LLMs Dethrone Traditional Approaches?

Comparison of Models: Other Participants

system	head-match
LLM-GLaRef-CRAC25	62.96
LLM-NUST-FewShot	61.74
LLM-PUXCRAC2025	60.09
LLM-UWB	59.84
CorPipeEnsemble	75.84
CorPipeBestDev	75.06
CorPipeSingle	74.75
Stanza	67.81
GLaRef-Propp	61.57
BASELINE-GZ	58.18
BASELINE	56.01

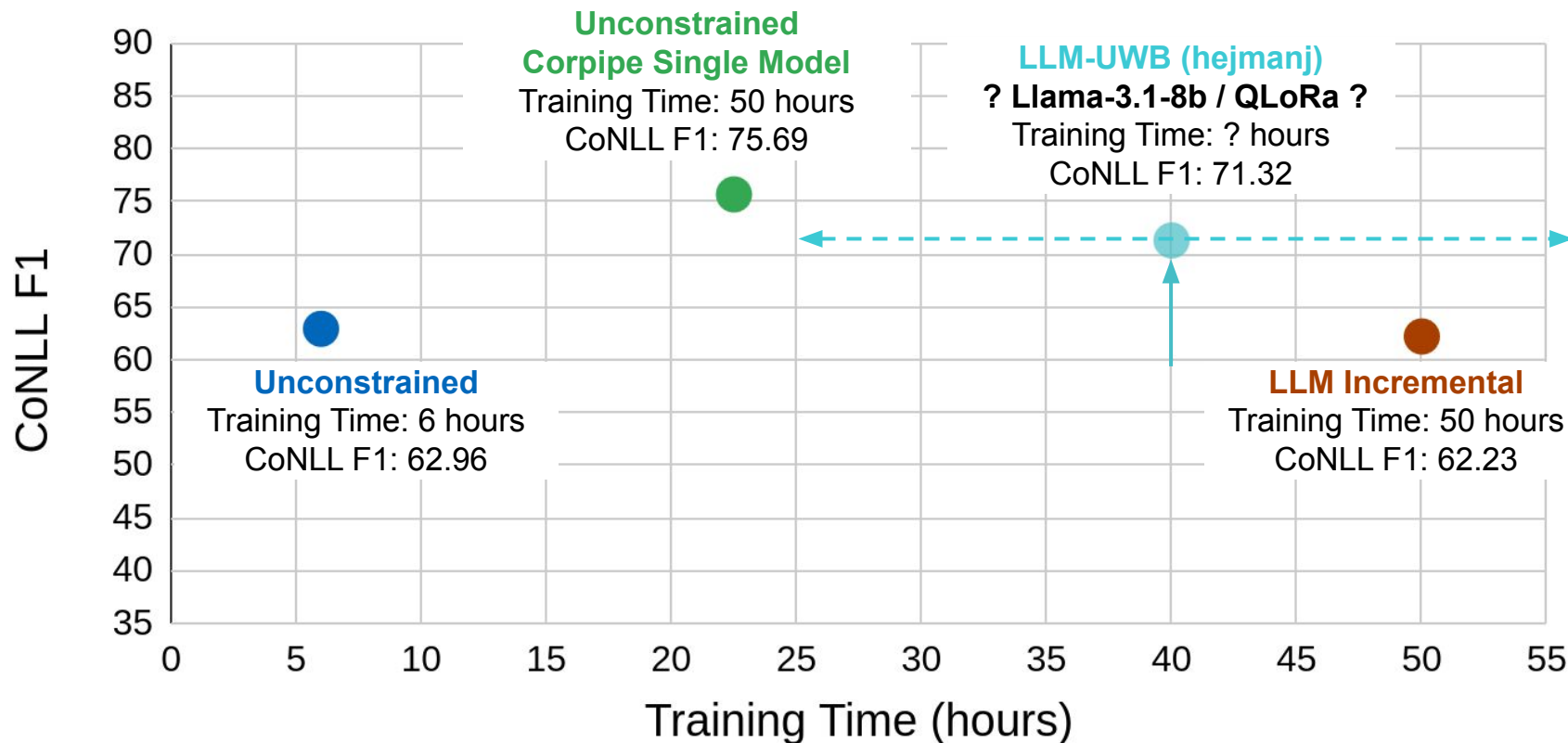
LLM-UWB (hejmanj) The UWB team fine-tunes a Llama-3.1-8B model on the official plain-text export of the CoNLL-U files. Training is done using QLoRA adaptation. The model is trained to

Findings of the Fourth Shared Task on Multilingual Coreference Resolution: Can LLMs Dethrone Traditional Approaches?

#	User	Entries	Date of Last Entry	avg ▲
1	hejmanj	5	07/29/25	71.32 (1)

Submitted after the official deadline

Comparison of Models: Other Participants

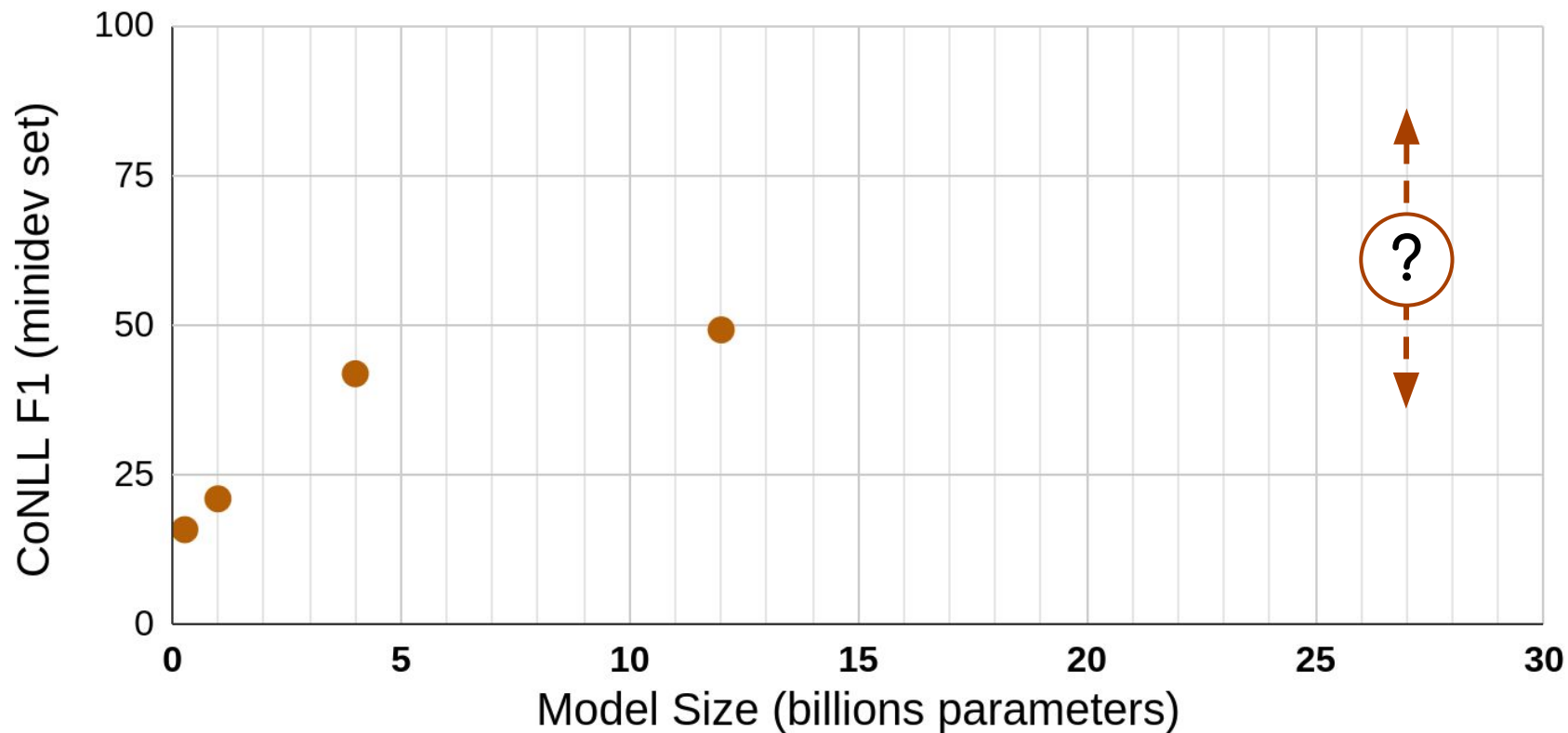


LLM Coreference Resolution: Perspectives

LLM Coreference Resolution: Perspectives

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format

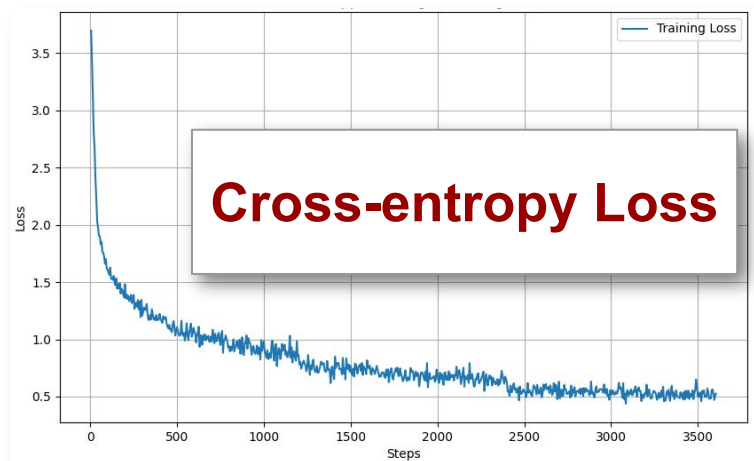
Perspectives: Model Size Increase



LLM Coreference Resolution: Perspectives

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format

Perspectives: Coreference-aware Loss Function



QLoRa
Fine-tuning

Plaintext

**Are incorrect coreference
IDs properly penalized?**

Annotated
CoNLL-U

Plaintext
to
CoNLL-U

So she|[e2] was considering in her|[e2] own mind (as
well as she|[e2] could , for the hot day made her|[e2]
feel very sleepy and stupid)

Fine-tuned

So she|[e2] was considering in her|[e2] own mind (as
well as she|[e3] could , for the hot day made her|[e2]
feel very sleepy and stupid)

Annotated

LLM Coreference Resolution: Perspectives

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format

Perspectives: Coreference ID Tracking

SYSTEM INSTRUCTION

PREVIOUS CONTEXT

TEXT INPUT

EXPECTED MODEL OUTPUT

<start_of_turn>user

You are a linguist, expert in anaphora and coreference resolution.
Based on the previous context, annotate in the input sentences which nouns, pronouns and other expressions refer to the same entity.
Do only insert annotations. Do not insert extra linguistic material, nor punctuation markers and do not delete elements from the input texts.

Previous context: *ANNOTATED SENTENCES FROM PREVIOUS BATCHES*

Maximum ID used: [e13]

Input: *PLAINTEXT SENTENCE BATCH*
<end_of_turn>

<start_of_turn>model

COREFERENCE ANNOTATED SENTENCE BATCH

<end_of_turn>

Perspectives: Coreference ID Tracking

SYSTEM INSTRUCTION

PREVIOUS CONTEXT

TEXT INPUT

EXPECTED MODEL OUTPUT

<start_of_turn>user

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Based on the previous context, annotate in the input sentences which nouns,
pronouns and other expressions refer to the same entity.
Do only insert annotations. Do not insert extra linguistic material, nor
punctuation markers and do not delete elements from the input texts.

Previous context: *ANNOTATED SENTENCES FROM PREVIOUS BATCHES*

Entity Tracker: "Alice's sister"[e1], "Alice"[e2], "The White Rabbit"[e3]

Input: *PLAINTEXT SENTENCE BATCH*
<end_of_turn>

<start_of_turn>model

COREFERENCE ANNOTATED SENTENCE BATCH

<end_of_turn>

LLM Coreference Resolution: Perspectives

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format

Perspectives: Coreference ID Tracking

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format

Perspectives: Plaintext format conversion

Down the|[e1 Rabbit-Hole|e1] Alice|[e2] was beginning to get very tired of sitting by her|[e2],[e3 sister|e3] on the |[e4 bank|e4] , and of having nothing to do : once or twice she|[e2] had peeped into the book her|[e2],[e3 sister|e3] was reading.

Down <e1>the Rabbit-Hole</e1> <e2>Alice</e2> was beginning to get very tired of sitting by <e3><e2>her</e2> sister</e3> on <e4>the bank</e4> , and of having nothing to do : once or twice <e2>she</e2> had peeped into thebook <e3><e2>her</e2> sister</e3> was reading.

Alternative tagging scheme inspired by markup languages like HTML or XML that tokenizers and LLMs might be more familiar with.

Perspectives: Coreference ID Tracking

1. Model Size Increase
2. Coreference-aware Loss Function
3. Coreference ID Tracking
4. Plaintext Format
- 5. Other Suggestions ?**

Thank you for your attention

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Additional Material: Incremental approach

PREVIOUS CONTEXT

RAW TEXT INPUT

EXPECTED MODEL OUTPUT

STEP 1

[None] CHAPTER I. Down the Rabbit-Hole Alice was beginning to get very tired of sitting by her sister on the bank , and of having nothing to do : once or twice she had peeped into the book her sister was reading , but it had no pictures or conversations in it , ' and what is the use of a book , ' thought Alice ' without pictures or conversations ? '

↓
Large Language Model
↓

CHAPTER I. Down the|[e1 Rabbit-Hole|[e1] Alice|[e2] was beginning to get very tired of sitting by her|[e2],[e3 sister|[e3] on the|[e4 bank|[e4] , and of having nothing to do : once or twice she|[e2] had peeped into the book her|[e2],[e3 sister|[e3] was reading , but it had no pictures or conversations in it , ' and what is the use of a book , ' thought Alice|[e2] ' without pictures or conversations ? '

Additional Material: Incremental approach

PREVIOUS CONTEXT

RAW TEXT INPUT

EXPECTED MODEL OUTPUT

STEP 2

CHAPTER I. Down the|[e1 Rabbit-Hole|e1] Alice|[e2] was beginning to get very tired of sitting by her|[e2],[e3 sister|e3] on the|[e4 bank|e4] , and of having nothing to do : once or twice she|[e2] had peeped into the book her|[e2],[e3 sister|e3] was reading , but it had no pictures or conversations in it , ' and what is the use of a book , ' thought Alice|[e2] ' without pictures or conversations ? ' So she was considering in her own mind (as well as she could , for the hot day made her feel very sleepy and stupid) , whether the pleasure of making a daisy-chain would be worth the trouble of getting up and picking the daisies , when suddenly a White Rabbit with pink eyes ran close by her .

↓
Large Language Model
↓

So she|[e2] was considering in her|[e2] own mind (as well as she|[e2] could , for the hot day made her|[e2] feel very sleepy and stupid) , whether the pleasure of making a daisy-chain would be worth the trouble of getting up and picking the daisies , when suddenly a|[e5 White Rabbit with pink eyes|e5] ran close by her|[e2] .